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Quantum Autoencoders for Network Anomaly Detection: A Comparative Benchmark with Classical Methods

Gönül Altay [1,*] and Ecir Uğur Küçüksille [2]

[1] Energy Management Unit, Manisa Provincial Directorate of National Education, Ministry of National Education, 45020 Manisa, Türkiye
[2] Department of Computer Engineering, Suleyman Demirel University, 32260 Isparta, Türkiye; ecirkucuksille@sdu.edu.tr
* Correspondence: gonulaltayy@gmail.com

Abstract

The practical value of quantum machine learning for network anomaly detection remains unclear under controlled and comparable benchmarking conditions. This study evaluates a trash-qubit-based Quantum Autoencoder (QAE) against seven classical anomaly-detection methods on UNSW-NB15 using shared data partitions, preprocessing steps, model selection procedures, and threshold calibration protocols. The experimental design included five independent seeds, three training set sizes, two feature budgets, and 12 QAE architectures, together with low-FPR analysis, finite-shot and synthetic-noise evaluation, less-constrained classical reference settings, and a cross-dataset assessment on TON_IoT. The Angle-RY encoding with a Real-Amplitudes Ring ansatz at depth $L = 2$ achieved the highest mean validation-selection AUPRC. On UNSW-NB15, the QAE was competitive in AUPRC and AUROC but remained behind the strongest classical baselines in F1-score, and its attack recall declined sharply under low-FPR constraints. The TON_IoT results further showed that relative model performance depended on the dataset. Under finite-shot and synthetic-noise conditions, anomaly-score rankings were largely preserved, whereas threshold-based decisions were more sensitive. Overall, the QAE showed competitive continuous-score discrimination on UNSW-NB15, but achieving reliable detection at low false-alarm rates remains a key operational challenge.

Keywords: quantum autoencoder; quantum machine learning; network anomaly detection; UNSW-NB15; TON_IoT; low-FPR

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1. Introduction

The emergence of new attack types in modern computer networks and changes in traffic distributions over time make anomaly-based intrusion detection an active area of research. Attacks not previously observed in the training data, data and concept drift, and difficulties in generalizing across different network environments remain among the primary challenges in reliable intrusion detection [1]. The structure of the anomalies and the characteristics of the data can directly affect the performance of detection methods. Large-scale benchmark studies have shown that different types of anomalies exhibit different behaviors and that no single detection method is systematically superior under all conditions [2–4]. These findings indicate that, despite extensive research into machine learning and deep learning approaches, model performance can vary with the underlying data and anomaly structure.

Quantum Machine Learning (QML) is an emerging computational approach investigated for anomaly detection [5]. Hybrid classical–quantum models integrating parameterized quantum circuits into autoencoder architectures [6] and quantum autoencoder-based network anomaly detection frameworks [7] have been proposed. However, performance gains obtained on limited datasets, against a small number of classical comparators, or within narrow quantum configuration spaces cannot alone be considered evidence of a general quantum advantage. QML benchmark results can be substantially affected by dataset selection, data encoding, circuit and optimization hyperparameters, classical baseline configuration, and other experimental design choices [8]. Recent systematic evidence further emphasizes that comparative QML evaluations should use well-tuned classical baselines and account explicitly for computational resources and experimental conditions when interpreting claims of practical quantum advantage [9]. Furthermore, QML-based anomaly detection studies differ considerably in terms of data-encoding strategy, model architecture, qubit count, circuit depth, noise handling, and evaluation environment. Systematic reviews indicate that performance values reported for quantum and classical methods across different studies are not always directly comparable and that more comprehensive evaluations are needed [1]. In classical anomaly detection, different methods are also known to perform better under different data and anomaly structures [4]. Therefore, quantum models should be compared with strong classical methods representing different assumptions about anomalous behavior under shared data partitions, preprocessing, model selection, threshold calibration, and evaluation protocols.

This study examines the relative performance of a trash-qubit-based Quantum Autoencoder (QAE) for network anomaly detection within a controlled comparative framework. To avoid evaluating the QAE based on a single circuit design, different data-encoding, ansatz, and circuit-depth options are compared. Furthermore, considering the operational importance of false alarms in network security applications, the behavior of the QAE at low-FPR operating points is evaluated, and the sensitivity of results obtained under ideal quantum simulation to finite-shot measurement and controlled noise conditions is examined. To assess whether the observed performance patterns extend beyond the primary benchmark, the selected QAE architecture is also evaluated on the independent TON_IoT network-security dataset without further architecture selection.

The main contributions of this study are as follows:

The performance of a trash-qubit-based QAE is compared with classical methods representing different assumptions about anomalous behavior under shared experimental conditions.

The effects of data-encoding method, ansatz family, and circuit depth on QAE performance are examined, and the final architecture is selected from multiple quantum configurations.

The extent to which the performance of the selected QAE configuration under analytic simulation is preserved under finite-shot measurement and controlled noise conditions is evaluated.

The behavior of the QAE under low false-alarm targets is examined by evaluating changes in test FPR and attack-detection performance at these operating points.

The transferability of the QAE architecture selected on UNSW-NB15 is evaluated on the independent TON_IoT network-security dataset without secondary architecture selection, providing a cross-dataset assessment of its relative performance and operational behavior.

Accordingly, this study focuses on the following research questions:

- RQ1: How does the QAE perform relative to classical anomaly detection methods in terms of F1, AUPRC, and AUROC under shared experimental conditions?

- RQ2: How do data-encoding method, ansatz family, and circuit-depth choices affect QAE performance, and which configuration emerges as the final QAE architecture?
- RQ3: To what extent is the performance of the selected QAE configuration preserved under finite-shot measurement and controlled noise conditions relative to analytic simulation?
- RQ4: To what extent do operating points defined by low false-alarm targets transfer to the test data, and to what extent is attack-detection performance preserved under these conditions?
- RQ5: To what extent does the QAE architecture selected on UNSW-NB15 retain its performance characteristics when evaluated on the independent TON_IoT network-security dataset under the same experimental framework?

The remainder of this paper is structured as follows. Section 2 reviews related work in classical and quantum anomaly detection. Section 3 details the QAE architecture, dataset preprocessing, classical baseline models, and statistical evaluation protocols. Section 4 presents the experimental results, including QAE architecture selection, low-FPR performance, comparisons with classical methods, cross-dataset evaluation on TON_IoT, finite-shot and noise sensitivity, supplementary sensitivity analyses, and computational cost. Section 5 discusses the implications, operational limitations, and findings. Finally, Section 6 concludes the paper and outlines directions for future research.

2. Related Work

2.1. Classical and Reconstruction-Based Anomaly Detection

Classical anomaly detection methods define anomalous behavior based on different structural assumptions. Isolation Forest (IF) isolates instances using random partitions [10], whereas Local Outlier Factor (LOF) evaluates local density through neighborhood relationships [11]. One-Class Support Vector Machine (OCSVM) estimates the support of the nominal data distribution using a kernel-based decision function [12]. Histogram-based Outlier Score (HBOS) utilizes histogram-based density estimates under the assumption of feature independence [13]. Among more recent methods based on distribution and tail behavior, Copula-Based Outlier Detection (COPOD) utilizes empirical copula-based tail probabilities [14], whereas Empirical Cumulative Distribution-Based Outlier Detection (ECOD) aggregates tail probabilities derived from univariate empirical cumulative distribution functions [15]. In autoencoder-based approaches, the reconstruction error computed from a representation learned exclusively on nominal data can be used as an anomaly score [16].

Large-scale benchmark studies show that the relative performance of algorithms can vary depending on the dataset, anomaly type, and evaluation conditions. The ADBench benchmark evaluated 30 algorithms across 57 datasets under varied supervision and anomaly settings [3]. Similarly, Bouman et al. compared 33 unsupervised methods across 52 real-world datasets and demonstrated that different methods can perform better under local and global anomaly structures [4]. These findings support evaluating a new anomaly detection approach not against a single classical model, but against baselines representing different decision mechanisms and assumptions about anomalous behavior.

2.2. Quantum Autoencoders for Anomaly Detection

QML-based anomaly detection is a rapidly developing research area in which parameterized quantum circuits and quantum-state representations are used for data modeling [5]. As one of the earlier approaches in this field, Sakhnenko et al. proposed a hybrid architecture in which a parameterized quantum circuit was incorporated into the bottleneck layer of a classical autoencoder and reported improved performance in their experimental

setting [6]. In network security, Hdaib et al. developed three quantum autoencoder-based frameworks for network anomaly detection [7].

QAE architectures have also been evaluated in other domains. For example, Ngairangbam et al. applied a Variational Quantum Circuit (VQC)-based QAE to anomaly detection in high-energy physics, compared it with classical autoencoders, and validated the model on available quantum hardware [17]. Park et al. proposed the Variational Quantum One-Class Classifier (VQOCC), which does not require a decoder and is trained using only normal data, and examined the effects of data-encoding strategy, number of circuit layers, and latent-space dimension on model performance [18].

More recent studies have focused on the role of architectural parameters. Frehner and Stockinger evaluated different ansatz designs and circuit depths for time-series anomaly detection and showed that QAEs could outperform classical autoencoders, although their performance depended strongly on the selected ansatz [19]. Similarly, Duffy et al. proposed a task-specific circuit ansatz for unsupervised anomaly detection [20], while Araz and Spannowsky showed that data-encoding (embedding) strategy plays an important role in determining the model's representational capacity and anomaly-discrimination performance [21].

More recently, Cirillo et al. conducted a systematic benchmark of supervised QML models for network intrusion detection, evaluating Pegasos-QSVC, VQC, and a hybrid quantum–classical neural network on the ToN_IoT and NSL-KDD datasets under both ideal and IBM-derived noisy simulation conditions [22]. Their analysis emphasized the effects of circuit configuration, entanglement, and backend-specific noise characteristics on model robustness and computational efficiency.

Recent network-focused studies have addressed complementary aspects of quantum anomaly detection. Chaudhary et al. investigated a trash-qubit QAE within a federated-learning framework for IoT network anomaly detection, with the primary emphasis on distributed versus centralized quantum training [23]. Alalyan et al. examined QSVM- and VQC-based intrusion detection on NSL-KDD and UNSW-NB15, focusing on dimensionality-reduction methods, dataset subsetting, and class distributions rather than QAE-based one-class detection [24]. More directly related to quantum–classical autoencoder benchmarking, Salem et al. compared quantum and hybrid autoencoder models with classical AE, VAE, Isolation Forest, and One-Class SVM baselines on NSL-KDD and ECG5000, emphasizing predictive performance and parameter efficiency under constrained model budgets [25]. Together with earlier QAE studies examining network-security applications [7], ansatz sensitivity and hardware execution [19], and encoding-dependent anomaly-detection behavior [21], these works demonstrate that QAE performance depends strongly on both model design and evaluation setting.

Quantum annealing and related adiabatic approaches represent a distinct quantum-computing paradigm from the gate-based variational models considered in this study. In machine learning settings, quantum annealers have been used for sampling and training energy-based models such as restricted Boltzmann machines. Dixit et al. demonstrated quantum-annealing-based RBM training on cybersecurity data using D-Wave hardware, while Moro and Prati subsequently evaluated quantum RBMs for network anomaly detection on real-world cybersecurity datasets and examined their computational behavior relative to classical RBMs [26,27]. Beyond anomaly detection, quantum annealing has also been investigated for real-world combinatorial optimization: Osaba et al. developed a hybrid quantum-classical approach for a practical package-delivery routing problem, and Ciacco et al. formulated a two-level facility-location problem for quantum annealing using a QUBO representation [28,29]. These studies illustrate the broader applicability of annealing-based quantum computation, but they differ fundamentally from the gate-based

QAE examined here. The present QAE learns a continuous anomaly score through a parameterized quantum circuit trained on benign data, allowing data encoding, variational ansatz, circuit depth, finite-shot effects, and gate-level noise to be studied directly within a common circuit-model framework.

Taken together, previous studies have examined important individual aspects of quantum anomaly detection, including QAE-based network-security applications, ansatz and encoding sensitivity, quantum-hardware execution, parameter efficiency, federated learning, and alternative QML classifiers. The contribution of the present study lies primarily in bringing several complementary evaluation dimensions together within a controlled QAE–classical benchmarking framework rather than in proposing a new QAE principle. Specifically, the study compares a trash-qubit QAE with classical anomaly-detection methods under shared data partitions and evaluation procedures; evaluates 12 predefined QAE configurations and the stability of architecture selection; examines operational behavior at stringent low-FPR targets; distinguishes finite-shot and synthetic-noise effects from fixed-threshold decision robustness; and evaluates cross-dataset behavior by transferring the UNSW-NB15-selected QAE architecture to TON_IoT without secondary architecture selection. This design enables QAE performance to be assessed not only in terms of continuous anomaly-score discrimination, but also with respect to architecture sensitivity, false-alarm constraints, execution perturbations, computational cost, and dataset dependence.

3. Materials and Methods

3.1. Dataset and Experimental Design

The primary matched benchmark in this study was conducted on the UNSW-NB15 dataset, which served as the main dataset for model development, architecture selection, calibration, and comparative evaluation. Empirical evaluations were conducted using the officially released partitions of the UNSW-NB15 dataset [30], comprising the training set (UNSW_NB15_training-set.csv, 175,341 records) and test set (UNSW_NB15_testing-set.csv, 82,332 records). The UNSW-NB15 dataset contains nominal network traffic and nine distinct attack categories: Analysis, Backdoors, DoS, Exploits, Fuzzers, Generic, Reconnaissance, Shellcode, and Worms. In this study, these nine attack families were aggregated into a single anomaly class, where nominal traffic was labeled as 0 (benign) and attack traffic as 1 (anomaly). To prevent data leakage, training, validation, and calibration subsets were sampled only from the official training partition, whereas final evaluation data were drawn strictly from the test partition. Test samples were not used during model training, QAE architecture evaluation and selection, or decision-threshold calibration.

All models were trained using only benign traffic samples ($y = 0$). Due to the computational cost of statevector quantum circuit simulation and parameter optimization relative to classical baselines, a maximum pool of 2000 benign samples was allocated for each random seed. From this pool, nested training subsets of 1000, 1500, and 2000 samples were constructed, keeping quantum simulation runtimes feasible while enabling controlled analysis across different sample size regimes. Distinct from the training data, an independent benign-monitor subset of 250 nominal instances was reserved for early stopping and checkpoint selection. Model and architecture selection were conducted on a dedicated validation-selection subset, while decision-threshold calibration was performed on an independent validation-calibration subset, avoiding the reuse of evaluation samples.

Class proportions within the validation and test subsets were determined through stratified sampling to reflect the overall class distributions of their source partitions. Consistent with the empirical attack proportions of 68.06% in the official training partition and 55.06% in the test partition, both the validation-selection and validation-calibration subsets comprised 240 benign and 510 attack instances ($N = 750$, 68.00% anomaly ratio).

The final test set comprised 337 benign and 413 attack instances (N = 750, 55.07% anomaly ratio). During data preparation, metadata attributes including the record identifier id, binary ground-truth label label, and categorical identifier attack_cat were removed from the feature space. For each seed, the 750-sample test subset was independently sampled from the official UNSW-NB15 test partition while preserving the benign/attack composition of 337 benign and 413 attack samples. Attack samples were stratified by attack category to maintain a comparable attack-family distribution across seeds. The five seed-specific test subsets were mutually non-overlapping, yielding 3750 distinct test instances in total. The same seed-specific test subsets were used for the QAE and all classical baselines. In addition to the locked 750-sample primary evaluation subset, a post-selection representativeness analysis was carried out for the QAE on the full official UNSW-NB15 test partition, comprising 82,332 samples (37,000 benign and 45,332 attacks). The previously selected QAE architecture, fitted model parameters, preprocessing pipeline, and original Q99 thresholds were retained without retraining, architecture reselection, or threshold recalibration.

Intrusion-detection literature has shown that feature reduction can substantially affect model performance, making it useful to evaluate full-feature and reduced-feature conditions separately [31,32]. For this reason, an additional full-feature/larger-training classical reference analysis was conducted to examine whether the constrained training sizes and feature budgets imposed in the primary benchmark artificially limited the performance of the classical methods. In this analysis, across each of the five independent seeds, the classical models were retrained on 9950 benign training samples; in addition, 250 benign samples were reserved for monitoring, and a separate calibration set of 3125 samples (comprising 1000 benign and 2125 attack samples) was allocated for threshold determination. The d = 5 and d = 8 feature selection step from the primary benchmark was omitted; instead, all input features produced by the preprocessing pipeline were retained, yielding 59 to 61 features depending on the seed. The trained classical models were then evaluated on the complete official UNSW-NB15 test partition, namely 82,332 samples. This supplementary analysis does not replace the primary matched benchmark; it solely examines the performance classical models achieve under a separate reference condition when provided with larger benign training sets and all usable input features obtained after preprocessing.

To address the asymmetry in tuning effort between classical and quantum models—arising from evaluating 12 distinct circuit configurations for the QAE against a constrained hyperparameter search for the CAE in the primary benchmark-we conducted an additional controlled CAE tuning sensitivity analysis. Recent literature emphasizes that autoencoder-based anomaly detection can be sensitive to architectural depth, bottleneck dimensions, and optimization hyperparameters, noting particularly that learning rates and model capacity can substantially alter reconstruction behavior and subsequent anomaly-score distributions [33,34]. We therefore established a controlled search space comprising 12 predefined CAE configurations. These configurations systematically combined network capacity, latent/bottleneck design, and learning rate, with the baseline CAE setup from the primary benchmark included directly in this space as *hidden_multiplier* = 2, original_like latent scheme, and $LR = 10^{-3}$.

The experiment followed the same matched design as the primary benchmark, comprising 30 conditions across five seeds, three training sizes, and two feature budgets. Training each of the 12 candidate CAE configurations across these conditions resulted in 360 training runs. Every CAE was trained exclusively on benign training instances using Mean Squared Error (MSE) as the reconstruction objective. Learning to reconstruct normal traffic and using reconstruction error as the anomaly score remains standard practice in autoencoder-based anomaly detection, with recent 2025–2026 studies consistently relying on MSE or comparable reconstruction losses for this purpose [34,35].

Under every condition, hyperparameter selection was guided solely by reconstruction MSE on the dedicated benign-monitor split, completely isolated from attack labels, validation attack instances, and test ground truth. For each condition, we selected the configuration that achieved the lowest benign-monitor MSE, thereby keeping the tuning pipeline strictly independent of anomaly-discrimination labels. The selected models were subsequently evaluated on the complete official UNSW-NB15 test partition. To isolate the effects of architectural tuning from threshold-selection dynamics, we relied on threshold-independent AUPRC and AUROC as the primary evaluation metrics. Crucially, this evaluation was not intended to replace the locked primary CAE benchmark or introduce a post hoc optimized baseline; rather, it was designed as an independent sensitivity test to verify whether a broader hyperparameter exploration would alter the core conclusions drawn from the QAE–CAE comparison.

In the QAE architecture, individual input features were mapped directly to qubits via rotation gates, making the input dimensionality d equal to the required qubit register count N_{qubits} ($d = N_{qubits}$). Because increasing the qubit register expands the statevector space exponentially (2^d), two feature budgets of $d \in \{5, 8\}$ were defined to keep the computational load manageable and to analyze model behavior at different input dimensions. To ensure a fair and paired comparison, identical 5- and 8-dimensional feature subsets were also applied to all classical baselines. The benchmark was replicated across five independent random seeds (seeds 0–4), yielding 30 paired experimental conditions per algorithm (5 seeds $\times$ 3 training sizes $\times$ 2 feature budgets).

Baseline algorithms were selected based on differences in their underlying assumptions regarding anomalous behavior rather than class diversity in the dataset. The benchmark encompasses reconstruction error, empirical copula tail estimation, empirical CDF-based tail estimation, histogram density estimation, recursive spatial partitioning, local density estimation, and kernel-based one-class boundary estimation.

A controlled QAE architecture evaluation space was defined to evaluate the QAE across multiple circuit designs. Two data-encoding methods (Angle-RY and Dense Angle-RY-RZ), three variational ansatz families (Basic-RY Ring, Real-Amplitudes Ring, and Strongly Entangling Layers), and two circuit depths ($L \in \{1, 2\}$) were combined, yielding 12 distinct QAE configurations. The Basic-RY Ring ansatz used single-qubit R_y rotations with a closed-ring CNOT entangling pattern per layer, whereas Real-Amplitudes Ring employed an R_y-ring-CNOT-R_y sequence. For Strongly Entangling Layers, the standard PennyLane parametric template was deployed. Each of the 12 configurations was examined across all 30 experimental conditions, yielding 360 QAE evaluations. Final architecture selection was based on validation-selection AUPRC, independent of test partitions and decision thresholds.

An automated Quantum Architecture Search (QAS) procedure was not employed in this work. To maintain strict benchmark control and ensure reproducibility, we instead evaluated a discrete pool of 12 predefined candidate QAE architectures. Recent studies highlight that QAS methodologies introduce added methodological overhead, including search algorithm design, computational budgets, and optimization heuristics, with efficiency and scalability remaining open bottlenecks [36]. A gate-based variational QAE was used because the evaluated model compresses input states through parameterized quantum circuits into latent and trash subsystems, consistent with recent QAE anomaly-detection formulations [19].

Because random-seed variation can affect the reproducibility and stability of machine learning results [37], we conducted an additional architecture-selection stability analysis to determine whether the selected QAE architecture was overly contingent on a particular seed or experimental setting. No new models were trained; instead, the 360 validation

outcomes from the primary architecture evaluation were reanalyzed. Candidate architectures were ranked according to mean validation-selection AUPRC, and selection stability was systematically tested using a leave-one-seed-out (LOSO) stability analysis protocol. In each LOSO iteration, one seed was held out entirely, and the top-performing architecture was re-identified using the mean validation-selection AUPRC of the remaining four seeds, directly testing the sensitivity of the final selection to any single seed partition.

To complement this rank-stability analysis, performance margins between the top-ranked architecture and its closest competitors were evaluated using a seed-level paired Bayesian analysis. Given the small number of independent replications (n = 5), uncertainty was reported explicitly; related model-comparison work has likewise emphasized the importance of uncertainty quantification when models are similar or samples are small [38].

To evaluate cross-dataset architecture transferability, we employed the network traffic component of the TON_IoT [39] dataset as an independent external domain. Specifically, we used the train_test_network.csv file, which comprises 211,043 records across 44 columns; after excluding the target/metadata columns label and type, 42 input features remained. The TON_IoT data were processed using the same preprocessing and variance-based feature-selection pipeline as the UNSW-NB15 benchmark, with all preprocessing transformations and feature-ranking steps fitted exclusively on the corresponding benign training partition within each seed and training-size condition. The dataset encompasses 50,000 benign records and 161,043 attacks spanning nine distinct categories: backdoor, DDoS, DoS, injection, MITM, password, ransomware, scanning, and XSS. To prevent leakage from exact duplicate records, we constructed a deterministic approximately 70/30 binary-stratified master split at the duplicate-group level. Exact duplicate full rows were grouped using a 64-bit pandas content hash computed over all original TON_IoT columns, including label and type, with the number of hash groups verified against the number of exact unique full rows. Each duplicate group was assigned wholly to either the training or test pool, yielding 148,003 and 63,040 records, respectively, with no content group shared across partitions.

To preserve TON_IoT as an independent external evaluation domain, we retained the exact QAE architecture selected on UNSW-NB15 without secondary tuning. Trainable parameters were reset and optimized from scratch using only benign TON_IoT training data. Decision-threshold calibration followed the same Q99 protocol described in Section 3.5, using the TON_IoT calibration split and applying the selected threshold unchanged to the corresponding test partition. This design separates architecture transfer from parameter transfer; a comparable architecture-transfer strategy, in which an architecture selected on a source dataset is trained on a target dataset, has been used in recent neural-architecture-transfer work [40].

No additional hyperparameter selection was performed for the classical models on TON_IoT. The final classical model configurations selected through the validation-selection procedure on the primary UNSW-NB15 benchmark were retained unchanged, and each model was refitted using the corresponding benign TON_IoT training data. Thus, the TON_IoT data were not used to retune the classical model hyperparameters.

Across all random seeds, we preserved the experimental geometry established on UNSW-NB15: models were trained on 1000, 1500, and 2000 benign samples alongside a dedicated benign-monitor split, while model validation and threshold calibration relied on matched sets of 240 benign and 510 attack instances. Final evaluation was conducted on an independent 750-sample test set (337 benign and 413 attack records). The test set was not used for training, architecture selection, or threshold calibration.

3.2. Data Preprocessing and Feature Selection

All preprocessing transformations and feature-selection procedures were fitted exclusively on the corresponding nominal training partition for each seed and sample-size combination. Missing values in numerical features were imputed using the median, and feature scaling was performed using RobustScaler to limit the influence of outliers on the distribution. Missing values in categorical features were imputed using the most frequent category (mode) and mapped to numerical representations via OneHotEncoder (*handle_unknown* = "ignore"). All fitted transformation parameters were frozen and applied without refitting to the benign-monitor, validation-selection, validation-calibration, and test partitions.

Following transformation, features exhibiting zero variance across the nominal training set were removed. The remaining features were sorted in descending order of empirical variance computed strictly on the nominal training data; alphabetical ordering of feature names was used as a deterministic tie-breaker for identical variance values. The top five features were selected for the $d = 5$ condition, and the top eight features were selected for the $d = 8$ condition, ensuring that the 5-dimensional feature space is a direct subset of the 8-dimensional feature space ($\mathcal{F}_5 \subset \mathcal{F}_8$). For $d = 5$, the selected feature set was identical across all seed and training-size conditions: response_body_len, ackdat, sinpkt, sload, and tcprtt. For $d = 8$, the first seven selected features were also stable across all conditions: response_body_len, ackdat, sinpkt, sload, tcprtt, dttl, and synack. The eighth selected feature varied between sttl and djit across conditions. Neither validation nor test instances were used for variance ranking or feature selection. For the QAE models, RobustScaler-normalized input vectors were mapped to rotation angles via the nonlinear transformation:

$$\theta(x_i) = \pi tanh(x_i), \tag{1}$$

which maps the transformed inputs to rotation angles in the interval $(-\pi, \pi)$.

To determine whether QAE performance hinges on selecting the highest-variance transformed features from benign training data, we conducted an additional feature-representation sensitivity analysis. Specifically, we evaluated Principal Component Analysis (PCA) as an alternative to the original variance-based top-d selection. PCA is a well-established unsupervised dimensionality reduction technique that maps high-dimensional data into a lower-dimensional subspace ranked by explained variance, without relying on label annotations [41]. Dimensionality reduction is particularly relevant in QML because available qubit counts and circuit resources constrain the dimensionality that can be processed directly, and recent comparative work has shown that the choice of reduced representation can materially affect downstream quantum-model performance [42]. The utility of PCA and related dimensionality-reduction strategies in intrusion detection is likewise well documented in recent IDS literature [43].

In this analysis, PCA was fitted strictly on benign training samples for each seed and training-size condition; at no point were attack records, validation, calibration, or test splits exposed to the transformation. The original preprocessing pipeline was maintained up to the variance-filtering step, after which the feature space was projected onto d principal components ($d \in \{5, 8\}$).

The PCA-based representation was evaluated under the same matched experimental design as the primary benchmark. For each condition, the previously selected QAE architecture was reinitialized and trained from scratch on the benign PCA-transformed data, then evaluated on the complete official UNSW-NB15 test partition. To isolate feature-representation effects from threshold calibration, sensitivity was assessed using threshold-independent AUPRC and AUROC only. PCA results were compared with the corresponding variance-based QAE results under matched conditions, with the six conditions first

averaged within each seed before paired PCA-minus-variance differences were summarized across the five seeds.

3.3. Classical Baseline Models

A limited set of predefined candidate configurations governing core model behavior was evaluated for each classical baseline. All candidate models were assessed under the same validation-selection protocol as the QAE, using the seed-level mean AUPRC across the 30 paired experimental conditions as the selection criterion. A single global configuration was then retained for each baseline across all seeds, sample sizes, and feature budgets (Table 1).

Table 1. Validation candidate selection and final hyperparameter configurations of the classical baseline models.

Model	Validation-Selection Candidates (Mean AUPRC ± SD Across Seeds)	Final Configuration
CAE	LR 10^{-3}: 0.8589 ± 0.0271; LR 3×10^{-4}: 0.8235 ± 0.0243	5–10–2–10–5/8–16–4–16–8; LR = 10^{-3}; ReLU; linear output; Adam; MSE loss; batch size = 128; max epochs = 100; patience = 10
IF	100 trees: 0.8095 ± 0.0260; 300 trees: 0.8064 ± 0.0251	n_estimators = 100; max_samples = 256; contamination = "auto"; max_features = 1.0; bootstrap = False; random_state = seed
LOF	k = 35: 0.7356 ± 0.0288; k = 20: 0.7245 ± 0.0247	n_neighbors = 35; novelty = True; metric = "minkowski"; p = 2; contamination = "auto"
OCSVM	ν = 0.10: 0.6382 ± 0.0315; ν = 0.05: 0.5795 ± 0.0333	kernel = "rbf"; ν = 0.10; gamma = "scale"; tol = 0.001; max_iter = −1
HBOS	auto: 0.7711 ± 0.0214; 10 bins: 0.7674 ± 0.0163	n_bins = "auto"; α = 0.1; tol = 0.5; contamination = 0.10
ECOD	Single candidate: 0.7953 ± 0.0214	ECOD_DEFAULT; contamination = 0.10; n_jobs = −1
COPOD	Single candidate: 0.7628 ± 0.0166	COPOD_DEFAULT; contamination = 0.10; n_jobs = −1

For the Classical Autoencoder (CAE), the architecture remained fixed while varying only the learning rate to avoid confounding architectural and optimization effects. Shallow, symmetric encoder–decoder architectures designed for 5- and 8-feature inputs were structured to compress inputs into a lower-dimensional bottleneck representation before reconstruction. To examine the sensitivity of Adam optimization to different step sizes, two learning rates—one approximately one-third of the other ($LR \in \{10^{-3}, 3 \times 10^{-4}\}$)—were compared. The $LR = 10^{-3}$ configuration yielded the higher validation-selection AUPRC. The CAE was implemented using PyTorch (torch.nn.Module) and trained on nominal samples using Mean Squared Error (MSE) reconstruction loss [16]. The MSE of each instance served directly as its anomaly score. The training budget was capped at 100 epochs, with early stopping configured at a patience of 10 epochs on the benign-monitor subset using best-weight restoration to prevent unnecessary training when monitoring loss ceased to improve.

For IF, ensemble capacity was evaluated by comparing 100 versus 300 isolation trees, with the subsample size per tree fixed at 256 samples. This corresponds to the standard subsample size in scikit-learn's *max_samples* = "auto" behavior when the training set exceeds 256 samples [44], preventing subsample size from acting as a confounding variable. The 100-tree configuration was retained. The *random_state* parameter was tied to each experimental seed to ensure that stochastic tree construction remained reproducible within the paired design.

For LOF, because the neighborhood size directly determines the locality of the anomaly evaluation, the default neighborhood size ($k = 20$) was compared against a broader prede-

termined configuration (k = 35). The final LOF configuration used k = 35. The parameter *novelty* = True was used to allow the fitted LOF model to score unseen validation and test samples, as required by scikit-learn documentation for *score_samples* and out-of-sample inference [45].

For OCSVM, an RBF kernel with variance scaling (*gamma* = "scale") was held constant, while the ν parameter was evaluated across two predefined candidate values ($\nu \in \{0.05, 0.10\}$). This choice is motivated by the fact that ν directly controls the fundamental behavior of the one-class formulation by setting an upper bound on the training error rate and a lower bound on the fraction of support vectors [46]. Among the two candidate values, ν = 0.10 yielded the higher validation-selection AUPRC.

For HBOS, a fixed histogram resolution was compared against automated feature-wise resolution [13]. The *n_bins* = "auto" setting determines the appropriate bin count for each feature individually, avoiding the use of a single bin count across all features; in the PyOD implementation, this setting uses its automatic feature-wise bin-selection procedure [47]. Other HBOS parameters remained fixed, and automatic binning was retained for the final configuration.

ECOD and COPOD were evaluated in their standard parameter-free formulations. ECOD derives anomaly scores from tail probabilities based on empirical cumulative distribution functions (CDFs) [15], whereas COPOD derives anomaly scores from empirical copula-based tail probabilities [14]. In both models, *contamination* = 0.10 was retained strictly for PyOD's internal binary threshold and label generation [48,49]. Because decision thresholds in the main benchmark were calibrated independently on the validation-calibration subset, this internal parameter did not govern the final test decision threshold. The setting $n_jobs = -1$ was used solely as a computational setting to leverage all available CPU cores during fitting and scoring, without altering the underlying mathematical definition of the anomaly scores [48,49].

In addition to the primary baseline set, Deep Support Vector Data Description (DeepSVDD) and Elliptic Envelope (EE) were evaluated as supplementary baselines under identical data splits and evaluation protocols to examine sensitivity to baseline selection [50,51]. DeepSVDD was implemented as a custom one-class DeepSVDD model in PyTorch and trained on benign training instances [50]. The network employed a 5–10–2 encoder architecture for 5-feature inputs and an 8-16-4 encoder architecture for 8-feature inputs; all linear layers were implemented without bias terms and utilized ReLU activations, omitting batch normalization and dropout. Prior to DeepSVDD optimization, autoencoder pretraining based on the same encoder structure was conducted using the Adam optimizer ($LR = 10^{-3}$, $weight_decay = 10^{-6}$, $batch_size$ = 128, maximum 100 epochs). In the subsequent DeepSVDD training phase, learning rate candidates of $LR \in \{10^{-4}, 10^{-3}\}$ were compared. The final DeepSVDD configuration used $LR = 10^{-4}$. Remaining parameters were fixed at Adam optimizer, $weight_decay = 5 \times 10^{-7}$, $batch_size$ = 128, and a maximum of 100 epochs, with early stopping applied on the benign-monitor set (*patience* = 10, *best-weight* restoration). The hypersphere center was initialized as the mean of the initial benign training representations, regularized with ε = 0.1 for near-zero components, and held constant throughout training. The optimization objective minimized the mean squared Euclidean distance between the latent representations and the fixed center, with the squared Euclidean distance serving directly as the continuous anomaly score.

EE was trained on nominal training instances using the scikit-learn EllipticEnvelope implementation [51]. For robust covariance estimation, two predetermined candidates—*support_fraction* = None and *support_fraction* = 0.75—were evaluated. The final EE configuration used *support_fraction* = 0.75. Other parameters were fixed at *contamination* = 0.10,

assume_centered = False, and *store_precision* = True, with random_state tied to each experimental seed. Because EE does not rely on iterative gradient-based training, no optimizer, learning rate, epoch limit, or early stopping was applied. Lower raw *score_samples* values correspond to more anomalous instances in scikit-learn; therefore, anomaly scores were defined as—*score_samples*(X) to preserve the common convention across all models that higher scores indicate greater anomalousness.

3.4. Quantum Autoencoder Design

This study employed a pure, trash-qubit-based QAE without a classical decoder. The QAE aims to compress the information carried by an input quantum state into a lower-dimensional latent subsystem while steering the remaining trash qubits toward a predefined reference state, $|0\rangle^{\otimes n_{trash}}$ [52].

The qubit layout and partition between the latent and trash subsystems are detailed in Table 2 and remained fixed across all experimental conditions. The latent register size was set to approximately 40% of the total qubit count to establish a distinct compression bottleneck while preserving a subspace capable of representing the multidimensional structure of nominal network traffic. In the QAE literature, while maintaining a latent space smaller than the input space forms the primary compression mechanism, the appropriate compression ratio depends on the specific structure of the input states, and no single universal latent-qubit ratio has been established [52–54]. Accordingly, the 5-qubit circuit used 2 latent and 3 trash qubits, whereas the 8-qubit circuit used 3 latent and 5 trash qubits.

Table 2. QAE design space and fixed training configuration.

Component	**Setting/Specification**
Model	Trash-qubit QAE
Qubit Configuration	q = 5: 2 latent + 3 trash; q = 8: 3 latent + 5 trash
Input Transformation	$\tilde{x}_i = \pi \tanh(x_i)$
Encoding Candidates	Angle-RY; Dense Angle-RY-RZ
Encoding Gate Count	Angle-RY: q single-qubit rotations; Dense Angle RY–RZ: 2q single-qubit rotations
Ansatz Candidates	Basic-RY Ring; Real-Amplitudes Ring; Strongly Entangling Layers
Ansatz Parameter Count	Basic-RY Ring: L × q; Real-Amplitudes Ring: 2L × q; Strongly Entangling Layers: 3L × q
Circuit Depth (**L**)	$L \in \{1, 2\}$
Basic-RY Ring Structure	$R_y(\theta) \rightarrow \text{ring} - \text{CNOT}$ per layer
Real-Amplitudes Ring Structure	$R_y(\theta) \rightarrow \text{ring} - \text{CNOT} \rightarrow R_y(\theta)$ per layer
Strongly Entangling Layers Structure	Parameterized single-qubit rotations (R_z–R_y–R_z) → entangling gates per layer
Approximate Total Gate Count	Basic: g_{enc} + 2Lq; Real-Amplitudes: g_{enc} + 3Lq; Strongly Entangling: g_{enc} + 4Lq (where g_{enc} = q for Angle-RY, g_{enc} = 2q for Dense Angle RY–RZ)
Parameter Initialization	$\theta \sim \mathcal{N}(0, 0.01^2)$
Optimizer	Adam; Learning rate = 10^{-3}; Weight decay = 0
Batch Size	Training: 4; Evaluation/Inference: 16
Early Stopping	Min. 5, Max. 30 epochs; Patience = 3; MinΔ = 10^{-4}; Evaluated on benign-monitor; Best weights restored

Table 2. *Cont.*

Component	Setting/Specification
Quantum Backend	PennyLane default.qubit (statevector); shots = None; diff_method = "backprop"
Architecture Selection Criterion	Validation-selection AUPRC; Global average with equal weighting across seeds
Tie-Break Rule	Fewer trainable parameters → Lower circuit depth → Deterministic configuration ID

The QAE design space was structured across three modular components, with evaluated options and circuit properties summarized in Table 2:

- Data Encoding: Encoding alternatives were chosen to compare single-axis R_y rotation (Angle-RY) against dual-axis R_y-R_z rotation (Dense Angle-RY-RZ), which represents the same feature across two independent rotation axes. Data encoding can alter the theoretical expressibility of parameterized quantum models [55], while the embedding strategy can affect representational capacity and anomaly detection performance in QAE-based settings [21].
- Ansatz Families: Different ansatz families (Basic-RY Ring, Real-Amplitudes Ring, and Strongly Entangling Layers) were evaluated to systematically compare the effects of parameter capacity and entanglement structures on anomaly discrimination. Ansatz design involves a problem-dependent trade-off among expressibility, trainability, circuit depth, and hardware compatibility [56]. In QAE-based anomaly detection specifically, performance has been shown to depend strongly on ansatz choice [19], while task-specific ansatz design has also been explored [20].
- Circuit Depth: Circuit depths were kept shallow ($L \in \{1, 2\}$) to evaluate the expressive power of an additional variational layer while keeping both quantum resource requirements and simulation computational cost within controlled limits.

The model was trained by minimizing a local cost function derived solely from the Pauli-Z expectation values of the trash qubits. This approach, measuring the proximity of trash qubits to the $|0\rangle$ reference state via Pauli-Z expectation values, aligns with the local cost formulation proposed for QAE-based one-class classification [18]. For an input sample x, this cost and anomaly score are defined as:

$$\mathcal{L}(x) = \frac{1}{2|\mathcal{T}|}\sum_{i\in\mathcal{T}}(1 - \langle Z_i\rangle), \tag{2}$$

where $\mathcal{T}$ denotes the set of trash qubits and $\langle Z_i\rangle$ represents the expectation value of the Pauli-Z operator on the i-th trash qubit. Minimizing this cost on nominal (benign) samples encourages the trash qubits to approach the $|0\rangle$ state ($\langle Z_i\rangle \to 1$), whereas higher scores reflect lower conformity to the learned nominal compression structure (anomalousness).

Optimizer, training, early-stopping, and quantum simulation settings were kept identical across architectures, so optimization settings did not introduce an additional comparison dimension. Early stopping was based only on the benign-monitor subset, so training duration was not influenced by attack labels. A minimum training period of 5 epochs, a patience of 3 epochs, and a minimum improvement threshold of 10^{-4} were used, with the best-performing weights restored at the end of training. The primary experiments used analytic statevector simulation (*shots* = None, *diff_method* = "backprop"), excluding finite-sampling variance and shot noise from the architecture comparison. To monitor training dynamics, parameter gradients were concatenated into a single vector prior to each mini-batch update. The gradient L_2-norm was computed as the Euclidean norm of this vector, and gradient variance was calculated across parameter gradients; epoch-level values were obtained by averaging the corresponding mini-batch results.

Global architecture selection was based on threshold-independent validation-selection AUPRC. Within each seed, validation-selection AUPRC was first averaged across the six experimental conditions, and the resulting seed-level values were then weighted equally. Test results were excluded from architecture selection. Architecture selection was therefore kept separate from decision-threshold calibration.

3.5. Model Selection and Decision-Threshold Calibration

Model selection and decision-threshold calibration were conducted on separate data partitions. Global model and QAE architecture selection used the validation-selection subset, whereas decision thresholds were calibrated on the independent validation-calibration partition. This separation was used to limit selection bias and overly optimistic performance estimates arising from repeated use of the same finite validation sample for sequential optimization and hyperparameter decisions [57]. Although model training was restricted to benign samples, threshold calibration was label-informed because calibration F1-score was computed using both benign and attack samples in the validation-calibration partition.

AUPRC was used as the primary criterion for global model and architecture selection. AUPRC evaluates the ranking quality of continuous anomaly scores independently of any decision threshold and summarizes the precision–recall trade-off for the anomaly class.

Raw anomaly scores were retained on their native numerical scales; only score orientation was standardized so that higher values indicated greater anomalousness. For IF, LOF, and OCSVM, the sign of the *score_samples* output was inverted; for CAE, the reconstruction MSE was used directly, as were the corresponding decision-function outputs for HBOS, ECOD, and COPOD. Candidate thresholds were derived from quantile positions of the validation-calibration score distribution. A fixed grid of 99 evenly spaced quantiles from 0.01 to 0.99 was used to limit sensitivity to differences in score scale and extreme observations while providing continuous- and discrete-scoring methods with the same candidate budget. The 99-point grid was a controlled experimental design choice rather than a literature-derived optimum. To determine whether the calibrated operating point was sensitive to the resolution of the predefined Q99 candidate grid, we conducted an additional threshold-grid sensitivity analysis. This evaluation required no model retraining, as it reused the existing calibration anomaly scores. The primary Q99 grid was compared with a coarser Q49 grid comprising every second quantile of the primary grid ($q \in \{0.02, 0.04, \ldots, 0.98\}$; 49 quantile levels), as well as a UNIQUE strategy that evaluated every distinct calibration score as a candidate threshold. Across all three approaches, final thresholds were chosen using the identical F1-maximization criterion and deterministic tie-breaking rules detailed below, and then applied without modification to the corresponding test sets.

Among candidate thresholds, the operating point that maximized calibration F1-score was selected as the final decision threshold. As the harmonic mean of precision and recall, the F1-score balances both dimensions when defining an operational decision point [58]. If multiple candidate thresholds produced the same maximum F1-score, a deterministic tie-breaking hierarchy based on operational security constraints was applied:

- Higher Precision: Priority was given to the operating point producing fewer false positives at the same F1 level.
- Lower FPR: If precision values were also identical, the threshold producing the lower false-positive rate on nominal traffic was preferred.
- Higher Numerical Threshold: If all preceding criteria remained tied, the higher numerical threshold—representing the most conservative criterion for declaring an anomaly—was selected as the final deterministic tie-breaker.

Quantiles were computed using linear interpolation consistently across all models, and duplicate threshold candidates were removed.

3.6. Low-FPR Operating-Point Calibration

In network anomaly detection, the decision threshold directly determines the operational trade-off between attack detection (recall) and the rate at which nominal traffic is incorrectly flagged as anomalous (False Positive Rate, FPR). Problems where false-positive errors are prioritized and constrained are addressed within the Neyman–Pearson framework in the statistical classification literature [59]. Under this framework, the probability of misclassifying a nominal class ($y = 0$) sample as an attack ($y = 1$) is bounded by a predefined α level, while the error for the opposing class (missed attacks) is minimized. For score-based models, deriving decision thresholds from order statistics of nominal samples is grounded in this theoretical foundation [59]. Recent work on adaptive anomaly thresholding emphasizes explicit control of false-positive and false-negative rates in large-scale systems, including network intrusion detection [60]. In operational security monitoring, high alert volumes and false positives can increase analyst workload and contribute to alert fatigue [61].

The low-FPR analysis in this study was designed as an empirical calibration procedure for comparing models at predefined low false-alarm operating points rather than as a formal Neyman–Pearson population error guarantee. The target FPR levels of 1%, 5%, and 10% ($r \in \{0.01, 0.05, 0.10\}$) were treated as three controlled operational scenarios with increasing tolerance, rather than as universal optimal operating points. For each target rate, the decision threshold was calculated using only the nominal anomaly scores in a separate benign-only low-FPR calibration set, without using attack labels or test data.

In each experimental condition, the benign-only low-FPR calibration set contained $N_{cal} = 5000$ nominal samples. For a given target FPR (r), the maximum allowable number of false positives (K) in the calibration set was formulated as:

$$K = \lfloor r \cdot N_{cal} \rfloor, \tag{3}$$

Accordingly, the maximum permitted numbers of false positives for the 1%, 5%, and 10% targets were $K = 50$, $K = 250$, and $K = 500$, respectively.

Threshold determination bypassed empirical candidate searching over F1, recall, or related classification metrics. Instead, nominal calibration scores were sorted in descending order to identify the boundary score immediately exceeding the K permitted false positives—specifically, the $(K + 1)$-th highest score. Because the decision rule flags any instance with an anomaly score greater than or equal to the threshold as an anomaly, placing the threshold exactly at this boundary risks letting tied nominal scores push the empirical false-positive rate beyond the target FPR. To preclude this edge case, the operational threshold was shifted strictly above the boundary score to the next representable floating-point value via the nextafter function:

$$\tau = nextafter(s_{(K+1)}, +\infty), \tag{4}$$

This adjustment excluded nominal samples tied at the boundary value and ensured that the empirical calibration FPR remained at or below the specified target. No further threshold tuning based on attack labels or classification performance was performed.

The resulting threshold was applied without modification to the complete official UNSW-NB15 test partition, which encompasses 37,000 benign and 45,332 attack instances. At this fixed operating point, we evaluated model performance across seven metrics: test FPR, recall, precision, F1-score, specificity, balanced accuracy, and the Matthews correlation coefficient (MCC). Importantly, the nominal rates of 1%, 5%, and 10% served strictly as

calibration targets rather than constraints enforced on the test set. Any divergence between the target and realized test FPR therefore reflects how reliably an operating point calibrated exclusively on benign data transfers to unseen test traffic.

Results were aggregated at the seed level. For each model, target FPR, and seed, metric values were averaged across the six experimental conditions, yielding five independent seed-level values per model–target pair. The mean and standard deviation across these five seeds are reported in the experimental results tables. Because models could exhibit different realized test FPRs at the same calibration target, test recall was not interpreted in isolation as a direct measure of superiority; recall and realized test FPR were evaluated jointly. To quantify binomial sampling uncertainty, 95% Wilson confidence intervals were calculated separately for realized FPR and recall from the corresponding condition-level integer counts. These intervals were kept distinct from seed-to-seed variability, which was summarized using the mean and sample standard deviation across the five independent seeds.

3.7. Evaluation Metrics and Statistical Analysis

Model performance was evaluated using three primary criteria: F1-score, AUPRC, and AUROC. The F1-score measures threshold-dependent operational performance obtained by applying the decision threshold calibrated on the validation-calibration subset to the test data. AUPRC and AUROC evaluate the discriminative ranking quality of continuous anomaly scores independently of any decision threshold. Precision–recall analysis further provides a direct characterization of the trade-off between precision and recall for the anomaly class [62].

The independent unit of replication for statistical inference was defined as the random seed. Each seed included six paired conditions spanning three training sizes ($N_{train} \in \{1000, 1500, 2000\}$) and two feature budgets ($d \in \{5, 8\}$). The 30 total conditions were not treated as independent observations to prevent pseudoreplication. Primary comparisons focused on the overall, equally weighted mean behavior across these six conditions rather than on local effects at a single training size or feature dimension. Therefore, the six experimental conditions were averaged within each seed per model and metric, yielding five independent seed-level observations per algorithm. All inferential statistics were conducted on these seed-level values.

A three-stage statistical analysis was used to assess whether performance differences were systematic across seeds:

- Testing for Systematic Differences Across Models (Friedman Test): To determine whether the eight models exhibited broadly comparable performance across independent seeds or whether certain models ranked systematically higher or lower, a separate Friedman omnibus test was conducted for each metric [63]. Because the Friedman test operates on within-seed ranks rather than absolute metric values, it provides a non-parametric framework for comparing multiple algorithms evaluated on the same data partitions.
- Small-Sample Reliability (Monte Carlo Permutation): Given the small number of independent seeds ($N_{seeds} = 5$), significance was assessed using a Monte Carlo permutation procedure rather than the asymptotic χ^2 approximation of the standard Friedman test. Model labels were randomly permuted within each seed block $B = 200{,}000$ times to construct the empirical null distribution under H_0 of no systematic performance differences among models. To avoid zero-valued Monte Carlo p-values, the corrected estimator was computed as:

$$p_{MC} = \frac{b + 1}{B + 1}, \tag{5}$$

where b denotes the number of permutations yielding a test statistic equal to or more extreme than the observed statistic [64].

- Multiple-Testing Control and Rank Concordance: Because three distinct performance metrics (F1-score, AUPRC, and AUROC) were evaluated simultaneously across the same model family, Holm's step-down procedure was applied to the three Friedman p-values to control the family-wise error rate [65]. The threshold for statistical significance was set at $\alpha = 0.05$. Finally, to measure how consistently model rankings were preserved across varying seeds rather than dispersing randomly, the omnibus effect size was evaluated using Kendall's coefficient of concordance (W).

Following the omnibus tests, a seed-level paired Bayesian analysis was conducted to compare the QAE with each classical baseline [66]. For each metric and baseline model, paired differences across the five independent seeds were defined as:

$$\delta_i = Metric_{QAE,i} - Metric_{Classical,i}, (i = 1, \ldots, 5), \quad (6)$$

With only five independent seeds ($n = 5$), the paired Bayesian analysis was used to characterize the magnitude, direction, and uncertainty of the differences beyond a binary significance decision. Paired differences were modeled using a normal likelihood with the reference prior $p(\mu, \sigma^2) \propto 1/\sigma^2$ [66]. Within this framework, the Bayesian formulation addresses three key questions:

- Magnitude and Uncertainty of Differences: The 95% credible interval (95% CrI) for μ defines the interval containing the mean performance difference with 95% posterior probability under the model and observed data. A credible interval lying entirely above zero indicates a favorable mean difference for the QAE under the model, whereas an interval containing zero reflects uncertainty in the direction or magnitude of the difference (with positive μ favoring the QAE and negative μ favoring the classical model).
- Posterior Probability of a Positive Mean Difference: The posterior probability $P(\mu > 0|data)$ quantifies the probability, under the model and observed data, that the mean QAE–classical difference is positive.
- Seed-Level Consistency and Sensitivity to Individual Seeds: Paired median differences and seed-direction counts-indicating how many of the five seeds favored the QAE versus the classical model-were evaluated to assess whether mean differences were disproportionately influenced by an extreme deviation in a single seed.

Because no universally accepted region of practical equivalence exists in network security, an artificial ROPE was not defined, and posterior distributions were evaluated directly without arbitrary threshold discretization. The paired Bayesian analysis was used for pairwise uncertainty quantification without introducing an additional family of frequentist pairwise null-hypothesis tests [66].

Given that the seed served as the fundamental replication unit, each comparison relied on only five paired observations. We therefore performed a prior-sensitivity analysis to ensure our Bayesian conclusions were not driven by prior specifications. Alongside the reference uninformative prior $p(\mu, \sigma^2) \propto 1/\sigma^2$, we evaluated two proper Normal–Inverse-Gamma conjugate priors centered at zero difference: a weakly informative formulation ($\kappa_0 = 0.10$, $\alpha_0 = 2$, $\beta_0 = 0.01$) and a more strongly centered variant ($\kappa_0 = 1.00$, $\alpha_0 = 2$, $\beta_0 = 0.01$). The posterior mean, 95% credible interval, and directional probability $P(\mu > 0|data)$ were calculated directly from closed-form Student-t solutions across all priors, precluding MCMC sampling and associated convergence concerns.

All experimental workflows were implemented in Google Colab using Python 3.12.13. Data preprocessing and classical baselines relied on NumPy 2.0.2, pandas 2.2.2, SciPy 1.16.3,

and scikit-learn 1.6.1, while QAE modeling and statevector simulations were conducted using PennyLane 0.45.1 and PyTorch 2.11.0. Post hoc inferential statistical tests were executed via scikit-posthocs 0.14.0. To maintain environmental parity, all quantum simulations and comparative classical benchmarks were executed strictly on CPU within the identical Colab runtime.

Computational cost was quantified using the training and inference times recorded during the benchmark runs. For each model, training time was defined as the wall-clock time required to complete model fitting, while inference time was defined as the time required to generate anomaly scores on the test partition. Condition-level times were first averaged across the training-size and feature-budget settings within each seed, and the resulting seed-level values were summarized as mean ± sample standard deviation across the five independent seeds. For the QAE, the reported runtimes reflect classical statevector simulation on PennyLane's default.qubit backend and should not be interpreted as execution times on physical quantum hardware.

3.8. Finite-Shot and Noise Evaluation Protocol

Primary QAE benchmark experiments used analytic statevector simulation on PennyLane's default.qubit backend (*shots* = None), providing a noiseless reference without finite-shot sampling variability. Pauli-Z expectation values were therefore obtained directly from the statevector rather than estimated from finite measurements.

The stability of the selected QAE architecture under finite-shot and synthetic noise conditions was examined through a separate sensitivity analysis. In physical quantum processors, expectation values are estimated from finite measurement shots, and quantum states are subject to environmental decoherence and gate noise [67,68]. Noisy quantum simulation provides a controlled setting for examining such effects, and local depolarizing models have been reported to approximate relevant error dynamics of quantum devices [68].

Given the computational cost of mixed-state simulation, sensitivity analysis was conducted on the largest training partition (N_{train} = 2000) across 5- and 8-qubit input conditions. Models were not retrained; frozen QAE weights obtained from the primary benchmark were used to evaluate finite-shot and synthetic noise effects during inference. Six experimental scenarios were configured:

1. Analytic simulation (shots = None, noiseless).
2. Noiseless 256 shots.
3. Noiseless 1024 shots.
4. Mild analytic noise (shots = None).
5. Moderate analytic noise (shots = None).
6. Mild noise + 1024 shots.

For the mild noise regime, single-qubit gate error was set to $p_1 = 0.001$, post-CNOT local error to $p_2 = 0.010$, and measurement error to $p_{meas} = 0.010$. In the moderate noise regime, these parameters were increased to $p_1 = 0.005$, $p_2 = 0.020$, and $p_{meas} = 0.020$, respectively. These error rates were not derived from calibration data for a specific quantum processor; they were defined as controlled synthetic stress-test levels to examine QAE sensitivity and robustness under increasing noise.

Noisy circuits were executed using PennyLane's default.mixed density-matrix simulator. Single-qubit symmetric depolarization was modeled using DepolarizingChannel, and measurement bit-flip error was modeled using BitFlip channels [68]. Noise channels were inserted at the same circuit locations across all noisy scenarios: a single-qubit depolarizing channel with probability p_1 was applied after each parameterized gate R_y, and a local depolarizing channel with probability p_2 was applied separately to each of the two inter-

acting qubits after every CNOT entangling gate. Accordingly, p_2 represents a controlled approximation of the local error introduced by a CNOT gate rather than a full two-qubit physical depolarizing process. Readout noise was modeled by applying a bit-flip channel with probability p_{meas} exclusively to the measured trash qubits immediately prior to Pauli-Z measurement.

The 256-shot, 1024-shot, and mild-noise + 1024-shot scenarios were each evaluated over three independent measurement repetitions to capture finite-sampling variability, whereas analytic scenarios were evaluated once because they were deterministic.

Under the condition-specific threshold recalibration policy, the standard Q99 protocol described in Section 3.5 was applied independently to the calibration scores generated for each noise and shot scenario. The resulting threshold was then applied to the test scores of that scenario. This allowed scenario-specific score shifts to be accommodated without using test labels.

As a complementary robustness check, a fixed-threshold transfer policy was evaluated without condition-specific threshold recalibration. For each seed and feature-budget condition, the Q99 threshold calibrated under ideal statevector execution was applied unchanged to the corresponding finite-shot and noisy test scores. F1-score and empirical FPR under the fixed-threshold policy were compared with the corresponding results obtained under condition-specific threshold recalibration.

Results were aggregated at the seed level. For conditions with finite shots, the three measurement repetitions were first averaged within each seed and feature budget. Results for $d = 5$ and $d = 8$ were then combined with equal weighting within each seed, yielding five independent seed-level observations per scenario; final means and standard deviations were computed across these values.

In addition to F1, AUPRC, AUROC, and FPR, the sensitivity analysis evaluated the Spearman rank correlation (ρ_s) between noisy and analytic simulation test anomaly scores, as well as the decision disagreement rate relative to analytic simulation classifications. This characterizes how finite shots and noise perturb individual instance rankings and operational classifications beyond macro-level classification metrics.

4. Results

4.1. Classical Baseline Performance

The results show that the effect of using a larger training set and a broader feature space varied considerably across the classical methods and depended on the performance metric considered. IF and HBOS benefited most clearly from the less-constrained setting, particularly in terms of the ranking-based metrics AUPRC and AUROC (Figure 1a,b). The improvement was more limited for CAE, whereas LOF remained comparatively strong under both evaluation settings across F1-score, AUPRC, AUROC, and the other decision-based metrics (Figure 1a–f).

In contrast, ECOD and COPOD showed a decline in performance under the less-constrained setting, especially for F1-score, AUPRC, and AUROC (Figure 1a–c). OCSVM exhibited a more mixed pattern across metrics: although some measures improved, the changes in Recall and FPR did not indicate a consistent gain in the same direction (Figure 1d,f). Considering all six metrics together therefore provides a more complete view of model behavior, capturing not only overall discrimination performance but also the balance between successful anomaly detection and false alarms.

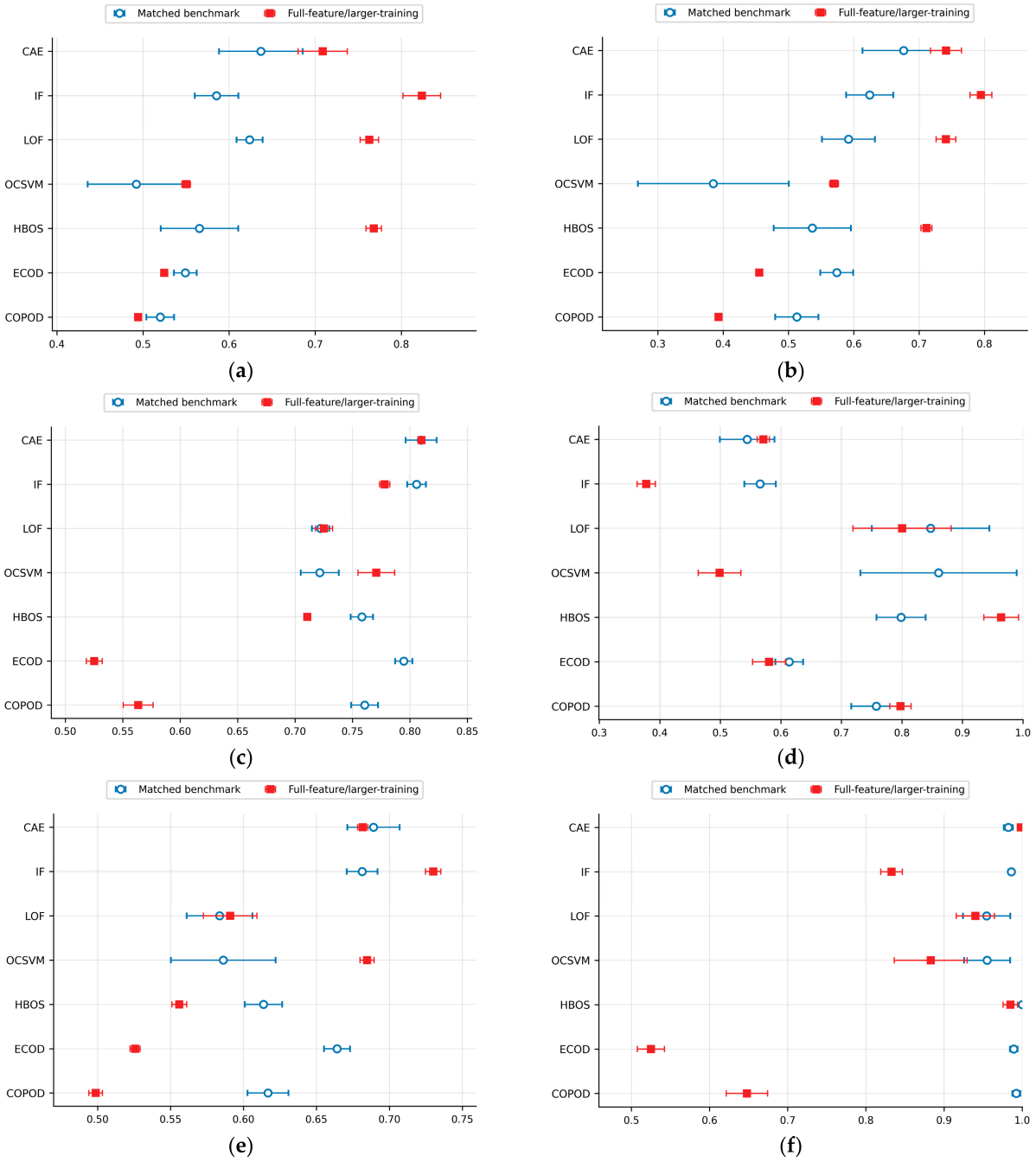


Figure 1. Performance comparison of classical anomaly detection methods under the matched benchmark and the less-constrained full-feature/larger-training setting. The figure reports (**a**) AUPRC, (**b**) AUROC, (**c**) F1-score, (**d**) False Positive Rate (FPR), (**e**) Precision and (**f**) Recall.

4.2. QAE Circuit-Design and Robustness Results

4.2.1. Circuit-Design Selection

Validation-selection rankings and metric distributions for the 12 evaluated QAE candidate architectures are summarized in Table 3. Analysis of the design space components reveals the following key empirical patterns:

Table 3. Validation-Selection Performance of Candidate QAE Circuit Designs.

Rank	Encoding	Ansatz	Depth	Validation-Selection AUPRC, Mean ± SD	Validation AUROC, Mean ± SD
1	Angle-RY	Real-Amplitudes Ring	2	0.7502 ± 0.0169	0.6779 ± 0.0267
2	Dense Angle-RY-RZ	Basic-RY Ring	2	0.7475 ± 0.0363	0.6775 ± 0.0524
3	Angle-RY	Strongly Entangling Layers	2	0.7471 ± 0.0540	0.6410 ± 0.0847
4	Dense Angle-RY-RZ	Real-Amplitudes Ring	2	0.7435 ± 0.0172	0.6629 ± 0.0241
5	Angle-RY	Basic-RY Ring	2	0.7418 ± 0.0224	0.6742 ± 0.0195
6	Dense Angle-RY-RZ	Strongly Entangling Layers	2	0.7374 ± 0.0400	0.6263 ± 0.0516
7	Dense Angle-RY-RZ	Strongly Entangling Layers	1	0.7108 ± 0.0111	0.6258 ± 0.0198
8	Dense Angle-RY-RZ	Real-Amplitudes Ring	1	0.7031 ± 0.0289	0.5840 ± 0.0337
9	Angle-RY	Basic-RY Ring	1	0.6568 ± 0.0335	0.5021 ± 0.0829
10	Angle-RY	Real-Amplitudes Ring	1	0.6567 ± 0.0217	0.4759 ± 0.0406
11	Angle-RY	Strongly Entangling Layers	1	0.6565 ± 0.0338	0.5017 ± 0.0836
12	Dense Angle-RY-RZ	Basic-RY Ring	1	0.6488 ± 0.0172	0.5037 ± 0.0286

The most pronounced separation was associated with circuit depth. All depth $L = 2$ architectures (ranks 1–6) occupied the upper half of the validation rankings, whereas all depth $L = 1$ configurations (ranks 7–12) fell into the lower half. Increasing the depth from 1 to 2 improved validation-selection AUPRC across all encoding–ansatz combinations. AUROC also increased in each matched comparison, although the magnitude of improvement varied across configurations.

At depth $L = 2$, Dense Angle-RY-RZ encoding yielded higher mean AUPRC than Angle-RY for the Real-Amplitudes Ring and Strongly Entangling Layers ansatzes, whereas the Basic-RY Ring showed a slightly higher AUPRC with Angle-RY encoding.

At depth $L = 2$, the top three architectures achieved closely matched AUPRC scores. Within this group, the Angle-RY + Real-Amplitudes Ring and Dense Angle-RY-RZ + Basic-RY Ring architectures attained the highest AUROC values.

Among the 12 candidates, the Angle-RY + Real-Amplitudes Ring + depth $L = 2$ configuration achieved the highest mean validation-selection AUPRC and was selected as the final QAE architecture. This architecture also achieved the highest mean validation AUROC, further supporting its continuous-score discrimination performance during validation.

The QAE architecture-selection stability analysis was conducted to assess how consistently the architecture with the highest mean validation-selection AUPRC remained preferable to its closest alternatives. The two nearest competing architectures achieved mean AUPRC values of 0.7475 and 0.7471, indicating only small differences in validation performance (Table 4). In the LOSO stability analysis, the selected architecture was chosen again in only 2 of the 5 evaluations, while the two closest competitors were selected in 2/5 and 1/5 evaluations, respectively. The paired Bayesian comparisons further showed that the 95% credible intervals for the differences between the selected architecture and these two competitors included zero (Table 5). Overall, these results show that the selected architecture ranked first in terms of mean validation-selection AUPRC, but did not demonstrate a clear or consistently stable advantage over the closest alternatives.

Table 4. Ranking and LOSO stability.

Architecture	Mean Validation-Selection AUPRC	Overall Rank	LOSO Selections
Angle-RY + Real-Amplitudes Ring, L = 2	0.7502	1	2/5
Dense Angle-RY-RZ + Basic-RY Ring, L = 2	0.7475	2	2/5
Angle-RY + Strongly Entangling Layers, L = 2	0.7471	3	1/5

Table 5. Paired Bayesian comparison with the selected architecture.

Comparator Architecture	Δ (Selected − Comparator)	95% CrI	P (Δ > 0)
Dense Angle-RY-RZ + Basic-RY Ring, L = 2	+0.00271	[−0.05733, 0.06275]	0.547
Angle-RY + Strongly Entangling Layers, L = 2	+0.00307	[−0.07461, 0.08075]	0.541

4.2.2. Training and Gradient Dynamics

Training dynamics and gradient stability of the final QAE architecture (Angle-RY + Real-Amplitudes Ring + depth $L = 2$) across the 30 experimental conditions are shown in Figure 2. Key observations across the training process include:

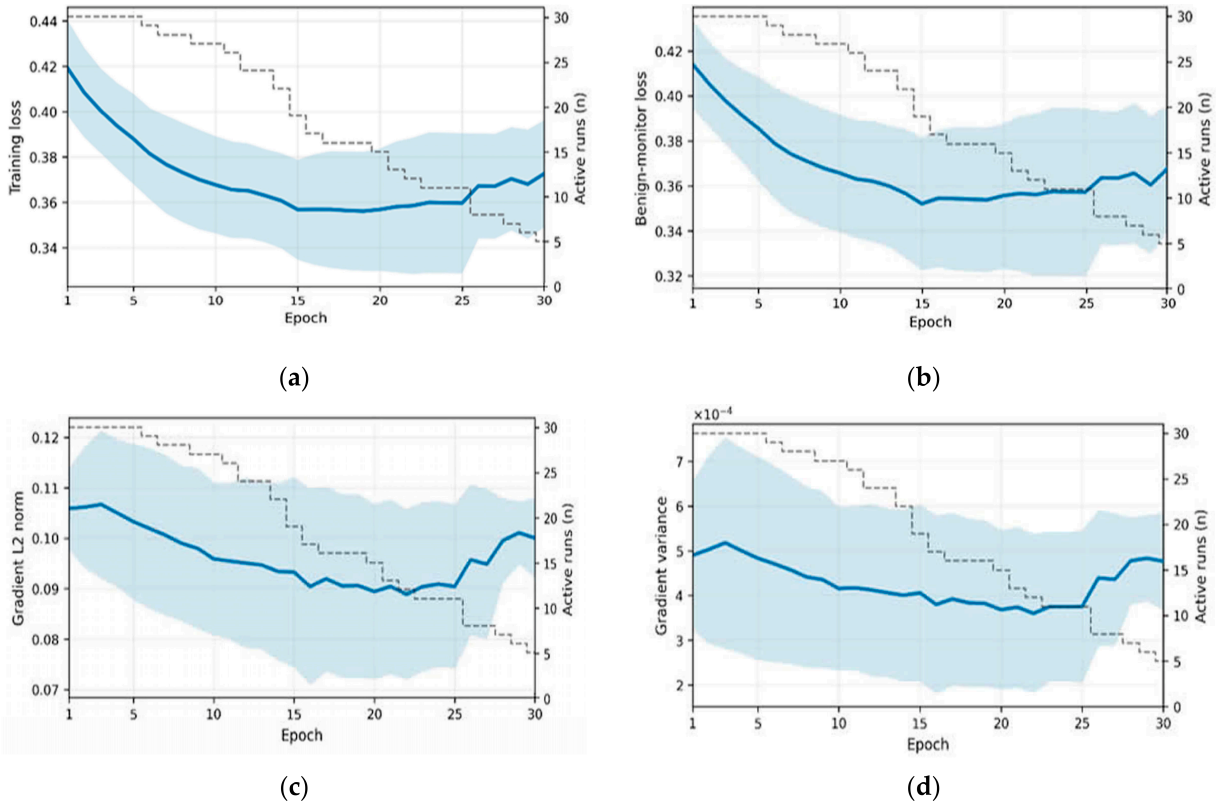


Figure 2. Training and gradient dynamics of the selected QAE architecture on UNSW-NB15: (**a**) Training loss, (**b**) Benign-monitor loss, (**c**) Gradient L_2 norm, and (**d**) Gradient variance. Solid lines represent the active-run mean, shaded areas indicate ±1 standard deviation, and dashed lines on the right axis denote the number of remaining active runs (n).

Training loss and benign-monitor loss decreased steadily over the first 15 epochs and entered a plateau between epochs 15 and 20 (Figure 2a,b). The median epoch at which the minimum benign-monitor loss was reached was 17 for the 5-feature conditions and 16 for the 8-feature conditions.

The number of active training runs decreased gradually after epoch 15 as early stopping terminated runs that no longer improved, with only 5 of the 30 runs continuing to the maximum budget of 30 epochs (Figure 2, dashed lines).

The gradient L_2-norm remained within approximately 0.03–0.08 and did not approach zero during training (Figure 2c). Parameter gradient variance exhibited a steady decrease from its peak in early epochs ($\sim 10^{-4}$) to intermediate epochs ($\sim 10^{-5}$) (Figure 2d). No clear indication of pronounced gradient explosion or vanishing-gradient/barren-plateau behavior was observed under the conditions examined.

The slight increase observed in mean loss and gradient curves after epoch 25 reflects the behavior of the small subset of remaining runs as the active run count declined ($N_{active} \leq 10$), rather than optimization instability (Figure 2a–d).

Overall, the selected QAE architecture converged within 15–20 epochs across most conditions while maintaining stable gradient behavior throughout training.

In the second-dataset evaluation, the QAE architecture selected on UNSW-NB15 was retained without any further architecture selection, while its trainable parameters were reinitialized and optimized from scratch using benign TON_IoT training data. This experiment was used to examine the training behavior of the selected architecture when transferred to a different dataset. On TON_IoT, the mean training loss decreased rapidly from approximately 0.37 to around 0.30 during the first few epochs, followed by a more gradual decline to approximately 0.27 around epochs 18–20 (Figure 3a). The benign-monitor loss followed a similar pattern, decreasing from about 0.37 to 0.30 and reaching approximately 0.28 around epochs 18–20 (Figure 3b).

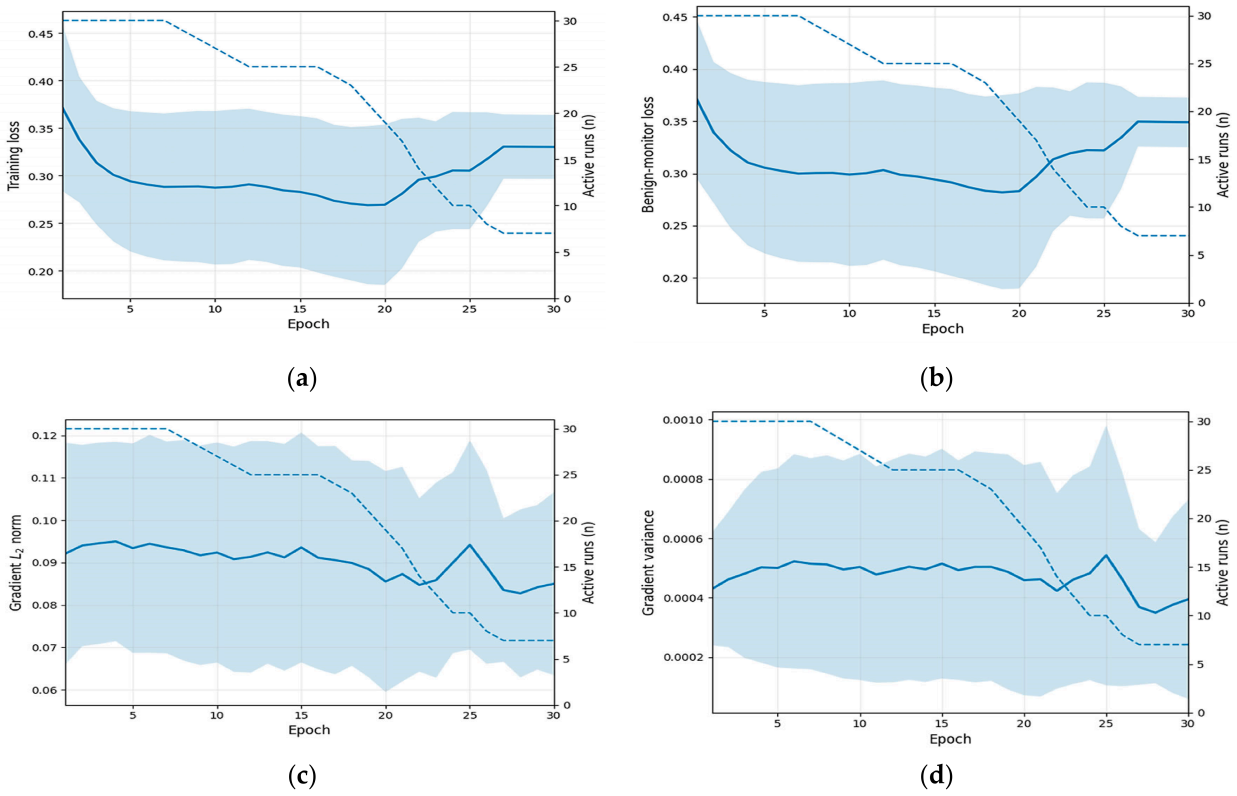


Figure 3. Training and gradient dynamics of the selected QAE architecture on TON_IoT: (**a**) Training loss, (**b**) Benign-monitor loss, (**c**) Gradient L_2 norm, and (**d**) Gradient variance. Solid lines represent the active-run mean, shaded areas indicate ± 1 standard deviation, and dashed lines on the right axis denote the number of remaining active runs (n).

As early stopping progressively removed completed runs, the number of active runs decreased from 30 at the beginning of training to about 25 around epoch 12, 10 around epoch 25, and 7 in the final epochs (Figure 3a,b, dashed lines). The gradient L_2 norm remained within an approximate range of 0.083–0.095 throughout training, with only limited variation across epochs (Figure 3c). Gradient variance remained mostly between approximately 4×10^{-4} and 5×10^{-4}, increased to around 5.4×10^{-4} near epoch 25, and then decreased to roughly 3.6×10^{-4}–3.9×10^{-4} in the final epochs (Figure 3d). Because of early stopping, averages at later epochs were calculated only from runs that remained active at the corresponding epoch.

4.2.3. Finite-Shot and Noise Robustness

The reliability of the QAE decision threshold established under ideal execution was examined under varying shot counts and noise conditions, and the results are presented in Table 6. Two thresholding strategies were compared. In the first, the decision threshold was recalibrated separately for each shot and noise condition using the corresponding calibration data. In the second, the threshold determined under ideal execution was kept fixed and applied unchanged to all other execution conditions.

Table 6. QAE performance under condition-specific threshold recalibration and fixed ideal-threshold transfer.

Scenario	F1 Recalibrated	F1 Fixed	FPR Recalibrated	FPR Fixed	AUPRC	AUROC
Analytic simulation	0.7543 ± 0.0098	0.7543 ± 0.0098	0.7576 ± 0.0310	0.7576 ± 0.0310	0.6180 ± 0.0367	0.6444 ± 0.0464
256 shots	0.7489 ± 0.0078	0.7490 ± 0.0087	0.7888 ± 0.0406	0.7181 ± 0.0259	0.6130 ± 0.0407	0.6453 ± 0.0470
1024 shots	0.7513 ± 0.0076	0.7502 ± 0.0106	0.7723 ± 0.0281	0.7330 ± 0.0283	0.6110 ± 0.0409	0.6450 ± 0.0474
Mild noise, analytic	0.7537 ± 0.0079	0.7398 ± 0.0046	0.7602 ± 0.0281	0.8608 ± 0.0252	0.6279 ± 0.0351	0.6599 ± 0.0411
Moderate noise, analytic	0.7542 ± 0.0084	0.7210 ± 0.0142	0.7540 ± 0.0289	0.9513 ± 0.0639	0.6374 ± 0.0384	0.6700 ± 0.0432
Mild noise + 1024 shots	0.7495 ± 0.0061	0.7407 ± 0.0052	0.7936 ± 0.0323	0.8575 ± 0.0267	0.6202 ± 0.0406	0.6580 ± 0.0441

When the threshold was recalibrated for each condition, the QAE showed relatively similar F1 and FPR results across the different execution settings. However, when the ideal threshold was transferred unchanged to noisy conditions, the FPR increased markedly, meaning that a larger number of benign samples were incorrectly classified as anomalies. In contrast, AUPRC and AUROC changed much less across the same conditions. This suggests that the main effect of noise was not a large loss in the model's ability to rank anomalous samples, but rather a reduced reliability of transferring a decision threshold calibrated under ideal execution directly to noisier settings.

Figure 4 shows the difference in FPR between recalibrating the threshold for each execution condition and keeping the ideal-condition threshold fixed. FPR remains relatively similar with recalibration, whereas the fixed ideal threshold produces a clear increase in FPR under noisy conditions.

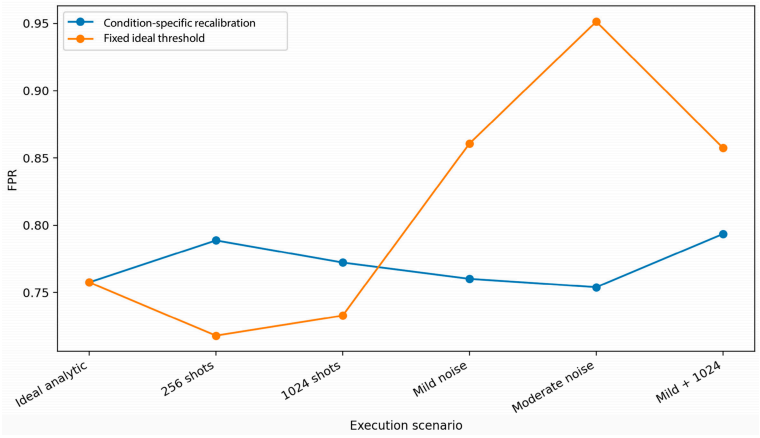


Figure 4. False positive rate under condition-specific threshold recalibration and fixed ideal-threshold transfer across finite-shot and synthetic-noise execution conditions.

Changes in score ranking and instance-level decisions were further examined using Spearman correlation and the decision disagreement rate (Appendix A, Table A10). Correlation with the ideal scores remained high across all execution conditions, with the 1024-shot setting showing higher rank agreement and lower decision disagreement than the 256-shot setting. Decision disagreement remained low under the analytic noise conditions,

while somewhat higher rates were observed for the finite-shot settings and the mild-noise + 1024-shot condition. Together with the F1 and FPR results reported in Table 6, these findings provide an additional view of how changes in the execution environment affect both anomaly-score ordering and individual classification decisions.

4.3. Low-FPR Operational Performance

Figure 5 shows how closely the nominal FPR targets of 1%, 5%, and 10%, set using benign-only calibration data, were maintained on the full UNSW-NB15 test set. This makes it possible to compare how well low-FPR operating points established during calibration transfer to unseen test data.

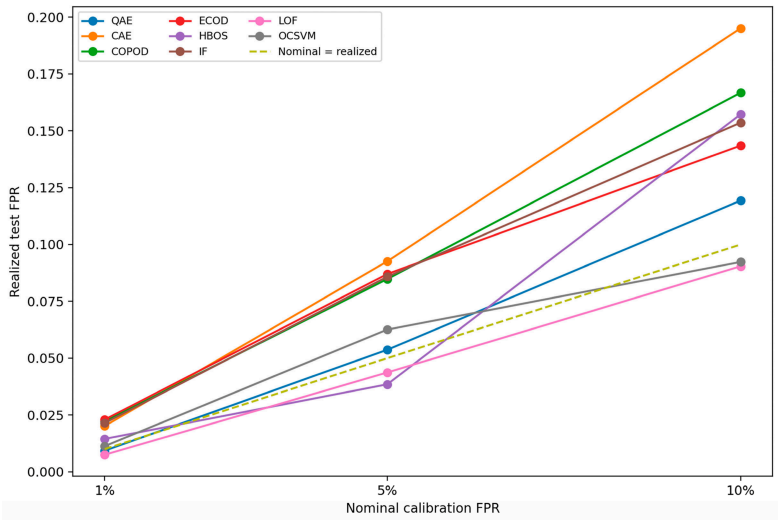


Figure 5. Transfer of nominal benign-only FPR targets to the full UNSW-NB15 test partition across anomaly detection models. Realized test FPR is shown for nominal calibration targets of 1%, 5%, and 10%. The dashed diagonal indicates perfect agreement between nominal and realized FPR.

The realized test FPR differed noticeably across models. For the QAE, realized FPR increased steadily as the nominal target increased and remained relatively close to the nominal line, particularly at the intermediate target. LOF and OCSVM also showed comparatively controlled transfer in the low-FPR region, whereas CAE, COPOD, ECOD, HBOS, and IF exhibited larger deviations, especially at higher nominal targets. Overall, the figure shows that low-FPR performance depends not only on the calibration target itself, but also on how well that target carries over to the test distribution.

DeepSVDD and EE, evaluated as supplementary baselines, were subjected to the same low-FPR calibration protocol (Appendix A, Table A4). DeepSVDD showed higher recall than the QAE at all three nominal operating points, although its realized FPR slightly exceeded the corresponding targets. EE showed lower recall than the QAE at the 1% and 5% targets but higher recall at 10%; its realized FPR remained below the nominal targets at 1% and 5% and exceeded the target at 10%. These results should be interpreted together with the realized FPR values, since recall comparisons at the same nominal target do not necessarily correspond to identical test FPRs. Additional threshold-dependent operational metrics, including F1-score, precision, specificity, balanced accuracy, and MCC, for all ten evaluated models are reported in Appendix A, Tables A5–A7.

The corresponding condition-level 95% Wilson confidence intervals for realized FPR and attack recall at the 1%, 5%, and 10% nominal targets are provided in Appendix A, Tables A8 and A9, respectively.

Figure 6 compares QAE and LOF specifically within the 0–10% FPR region. Across this range, LOF maintains higher recall than QAE, while the gap between the two models

becomes more visible as the allowable FPR increases. The shaded regions also show the variation across seeds, with QAE remaining at comparatively low recall levels throughout the evaluated low-FPR range.

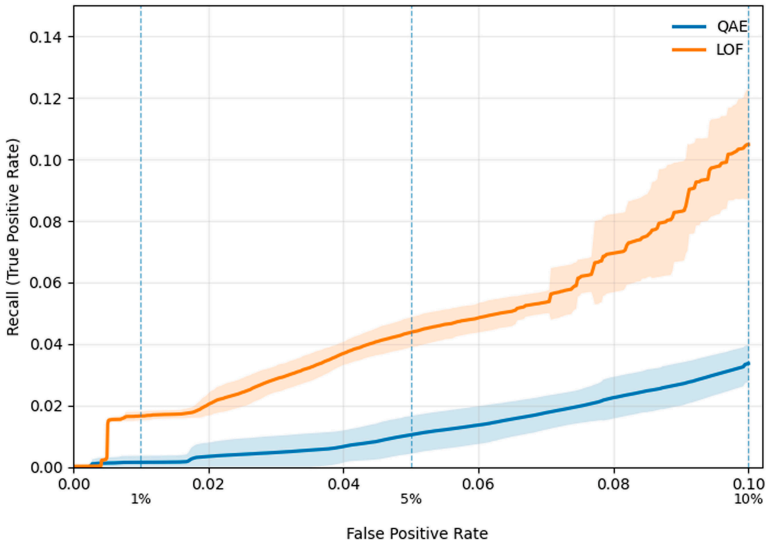


Figure 6. Low-FPR performance of QAE and LOF on the full UNSW-NB15 test partition. Recall is shown across the 0–10% false positive rate region. Solid lines represent the mean recall across five seeds, shaded areas indicate ±1 sample standard deviation, and vertical dashed lines mark the predefined nominal FPR targets.

4.4. Comparison of the QAE and Classical Models

The absolute performance values of the QAE and classical baselines on UNSW-NB15 are reported in Table 7, while the corresponding results on TON_IoT are presented in Table 8. The results show that relative model performance changed across datasets. On UNSW-NB15, CAE and IF achieved the strongest F1-scores, while the QAE remained competitive in AUPRC and AUROC. On TON_IoT, LOF achieved the strongest overall performance across F1-score, AUPRC, AUROC, precision, and FPR. The QAE maintained very high recall on TON_IoT, but this was accompanied by a substantially higher false-positive rate.

Table 7. Performance of the QAE and classical baseline models on UNSW-NB15.

Model	F1	AUPRC	AUROC	Precision	Recall	FPR
QAE	0.7576 ± 0.0088	0.6128 ± 0.0226	0.6408 ± 0.0266	0.6167 ± 0.0116	0.9847 ± 0.0056	0.7584 ± 0.0368
CAE	0.8098 ± 0.0134	0.6368 ± 0.0485	0.6762 ± 0.0628	0.6891 ± 0.0179	0.9822 ± 0.0058	0.5443 ± 0.0451
IF	0.8058 ± 0.0081	0.5854 ± 0.0254	0.6243 ± 0.0360	0.6813 ± 0.0105	0.9860 ± 0.0034	0.5657 ± 0.0259
LOF	0.7222 ± 0.0076	0.6238 ± 0.0151	0.5919 ± 0.0406	0.5836 ± 0.0225	0.9545 ± 0.0302	0.8475 ± 0.0971
OCSVM	0.7215 ± 0.0165	0.4922 ± 0.0566	0.3850 ± 0.1156	0.5861 ± 0.0359	0.9551 ± 0.0294	0.8605 ± 0.1291
HBOS	0.7582 ± 0.0097	0.5656 ± 0.0450	0.5365 ± 0.0592	0.6137 ± 0.0128	1.0000 ± 0.0000	0.7986 ± 0.0406
ECOD	0.7946 ± 0.0075	0.5491 ± 0.0132	0.5739 ± 0.0252	0.6641 ± 0.0089	0.9893 ± 0.0049	0.6138 ± 0.0229
COPOD	0.7606 ± 0.0117	0.5200 ± 0.0160	0.5127 ± 0.0331	0.6167 ± 0.0140	0.9925 ± 0.0051	0.7578 ± 0.0414

Table 8. Performance of the QAE and classical baseline models on TON_IoT.

Model	F1	AUPRC	AUROC	Precision	Recall	FPR
QAE	0.7296 ± 0.0076	0.6396 ± 0.0245	0.5603 ± 0.0185	0.5774 ± 0.0106	0.9960 ± 0.0065	0.9085 ± 0.0319
CAE	0.7480 ± 0.0185	0.6767 ± 0.0314	0.6348 ± 0.0508	0.6115 ± 0.0258	0.9802 ± 0.0270	0.7936 ± 0.0915
IF	0.7451 ± 0.0133	0.6754 ± 0.0216	0.6125 ± 0.0261	0.6014 ± 0.0257	0.9901 ± 0.0206	0.8272 ± 0.0865
LOF	0.8006 ± 0.0077	0.8469 ± 0.0155	0.8219 ± 0.0114	0.7588 ± 0.0428	0.8929 ± 0.0466	0.4227 ± 0.1037
OCSVM	0.7043 ± 0.0025	0.5439 ± 0.0305	0.4290 ± 0.0456	0.5481 ± 0.0014	0.9851 ± 0.0054	0.9952 ± 0.0032
HBOS	0.7102 ± 0.0000	0.5490 ± 0.0051	0.4724 ± 0.0169	0.5507 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
ECOD	0.7155 ± 0.0029	0.6663 ± 0.0225	0.5869 ± 0.0285	0.5571 ± 0.0036	1.0000 ± 0.0000	0.9752 ± 0.0134
COPOD	0.7102 ± 0.0000	0.6604 ± 0.0224	0.5665 ± 0.0288	0.5507 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

Supplementary evaluations with DeepSVDD and EE were conducted under the same matched experimental protocol, with detailed results provided in Appendix A, Table A1 for UNSW-NB15 and Appendix A, Table A2 for TON_IoT.

The relative performance of the QAE and classical models was compared across UNSW-NB15 and TON_IoT using average model ranks based on F1-score, AUPRC, and AUROC (Figure 7). The model rankings varied across both datasets and performance metrics. On UNSW-NB15, CAE and IF ranked among the strongest methods across several metrics, whereas on TON_IoT, LOF achieved the best rankings, particularly for AUPRC and AUROC. The QAE generally remained in the middle of the ranking across both datasets, although its relative position varied depending on the metric considered.

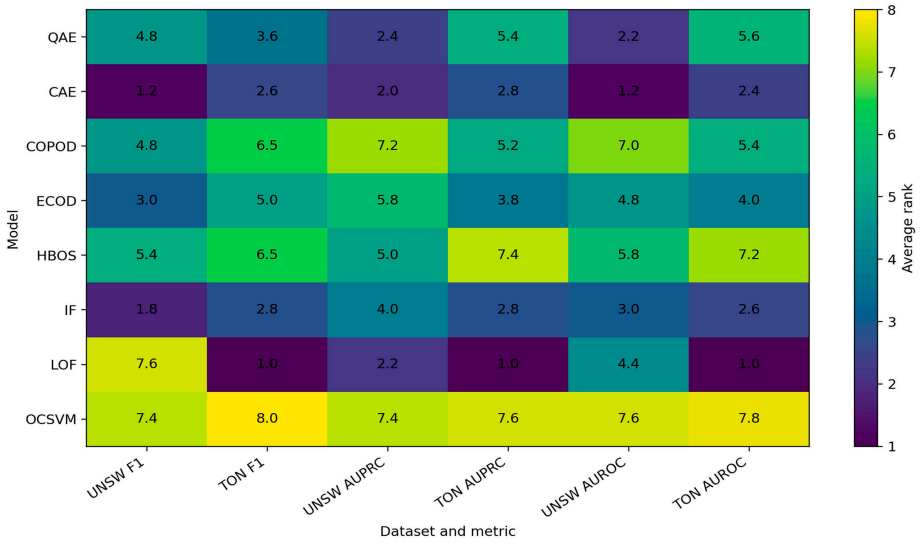


Figure 7. Cross-dataset average model ranks on UNSW-NB15 and TON_IoT. Average ranks are reported for F1-score, AUPRC, and AUROC, where a lower rank indicates better relative performance (1 = best).

Results of the Monte Carlo permutation Friedman omnibus tests, conducted to assess whether observed performance rank differences were systematic across seeds, are reported in Table 9. Following Holm step-down correction for multiple testing across the three metrics, the omnibus tests remained statistically significant ($p_{MC} < 0.05$) for F1-score, AUPRC, and AUROC (all Holm-adjusted $p < 0.0001$). The omnibus effect size, measured by Kendall's coefficient of concordance, was $W = 0.9390$ for F1-score, $W = 0.8590$ for AUROC, and $W = 0.8057$ for AUPRC, indicating a high degree of ranking consistency across independent seeds.

Table 9. Friedman omnibus test results and concordance coefficients across the eight evaluated models on UNSW-NB15.

Metric	Friedman Q	Monte Carlo p	Holm-Adjusted p	Kendall's W
F1	32.8667	<0.0001	<0.0001	0.9390
AUPRC	28.2000	<0.0001	<0.0001	0.8057
AUROC	30.0667	<0.0001	<0.0001	0.8590

Table 10 shows significant differences in model performance across all three evaluation metrics on TON_IoT. The Friedman omnibus tests remained significant after Holm correction for F1-score, AUPRC, and AUROC, indicating systematic performance differences among the eight evaluated models. The high Kendall's W values further indicate that the relative rankings of the models were broadly consistent across the five independent seeds.

Table 10. Friedman omnibus test results and concordance coefficients across the eight evaluated models on TON_IoT.

Metric	Friedman Q	Monte Carlo p	Holm-Adjusted p	Kendall's W
F1-score	33.7855	<0.0001	<0.0001	0.9653
AUPRC	31.5333	<0.0001	<0.0001	0.9010
AUROC	33.9333	<0.0001	<0.0001	0.9695

4.5. Paired QAE–Classical Analysis

Figure 8 presents the seed-level paired Bayesian differences between the QAE and each classical comparator. Positive values favor the QAE, whereas negative values favor the corresponding classical model. On UNSW-NB15, the QAE showed positive differences against OCSVM across all three metrics and also outperformed HBOS, ECOD, and COPOD in AUPRC and AUROC. In contrast, F1 differences against CAE and IF favored the classical models. The comparison with LOF was more metric-dependent, with the direction of the difference changing across F1, AUPRC, and AUROC (Figure 8a).

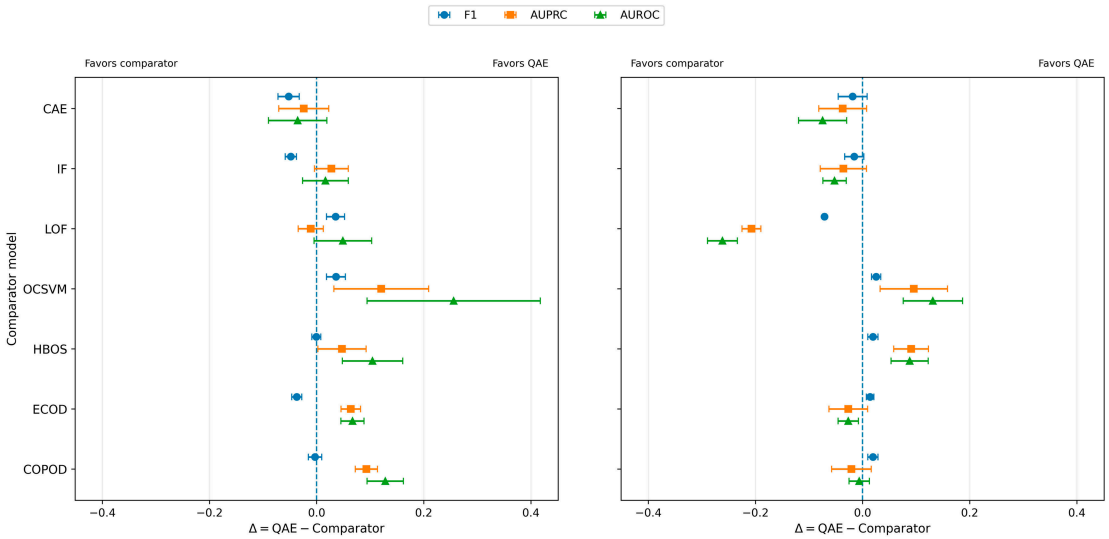


Figure 8. Paired Bayesian comparison of QAE and classical models on (**a**) UNSW-NB15 and (**b**) TON_IoT. Points indicate the posterior mean difference, defined as Δ = QAE − Comparator and horizontal error bars show the corresponding 95% credible intervals. Positive values favor the QAE, while negative values favor the classical comparator.

On TON_IoT, the separation between models was more pronounced. LOF outperformed the QAE across F1, AUPRC, and AUROC, whereas the QAE showed positive differences against OCSVM and HBOS on all three metrics. Comparisons with CAE and IF generally favored the classical models, while the results against ECOD and COPOD varied by metric (Figure 8b). In several comparisons, the 95% credible intervals crossed zero, whereas the more pronounced differences remained entirely on one side of zero.

Supplementary paired comparisons with DeepSVDD and EE are reported in Appendix A, Table A3. The QAE showed higher F1 and AUROC than DeepSVDD across all five seeds, while the AUPRC difference was more uncertain. In comparison with EE, the QAE showed lower F1, a small and uncertain advantage in AUPRC, and essentially no consistent difference in AUROC.

4.6. Sensitivity Analyses

4.6.1. CAE Tuning Sensitivity

The broader CAE tuning experiment produced only small changes relative to the original CAE configuration (Table 11). AUPRC increased slightly on average, but the 95% credible interval included zero, while AUROC showed almost no mean change. The posterior probabilities and seed-level directions were also mixed, indicating that the broader tuning search did not materially alter the overall CAE performance pattern observed in the primary benchmark.

Table 11. Sensitivity of CAE performance to broader architecture tuning.

Metric	Mean Δ Tuned − Original	95% CrI	P (Δ > 0)	Positive Seeds
AUPRC	+0.00577	[−0.01046, 0.02200]	0.810	4/5
AUROC	−0.00031	[−0.01967, 0.01904]	0.483	3/5

4.6.2. PCA Feature-Representation Sensitivity

Table 12 presents the sensitivity of QAE performance to the feature-representation strategy. Replacing the variance-based feature selection with PCA resulted in lower mean AUPRC and AUROC values across the five seeds. The reduction was more pronounced for AUROC, whose 95% credible interval remained below zero, whereas the AUPRC interval still included zero. Positive PCA-minus-variance differences were observed in only one of the five seeds for both metrics. Overall, the results indicate that the selected QAE performed better with the original variance-based representation than with the PCA-based alternative under the matched evaluation conditions.

Table 12. Sensitivity of QAE performance to PCA-based feature representation.

Metric	Mean Δ (PCA − Variance)	95% CrI	P(Δ > 0)	Positive Seeds
AUPRC	−0.04138	[−0.10019, 0.01743]	0.061	1/5
AUROC	−0.10560	[−0.19722, −0.01399]	0.016	1/5

4.6.3. Threshold Sensitivity Analysis

Because decision-threshold selection directly shapes operational classification metrics [58], sensitivity to the 99-quantile grid (Q99) protocol used in the primary benchmark was evaluated against a coarser grid, Q49 ($q \in \{0.02, 0.04, \ldots, 0.98\}$, 49 quantiles), and a UNIQUE strategy that treats all unique calibration scores as threshold candidates.

As shown in Table 13, core QAE performance remained largely consistent across threshold-selection strategies, with only minor variations in precision, recall, and FPR. Because AUPRC and AUROC are threshold-independent measures of continuous-score ranking quality [62], these metrics remained unchanged across Q99, Q49, and UNIQUE. Variations in threshold-dependent metrics such as F1-score and FPR alongside invariant AUPRC and AUROC are therefore expected when the threshold-selection strategy is changed.

Table 13. Threshold-Grid Sensitivity of the Selected QAE Architecture.

Threshold Method	F1	AUPRC	AUROC	Precision	Recall	FPR
Q99 (main)	0.7576 ± 0.0088	0.6128 ± 0.0226	0.6408 ± 0.0266	0.6167 ± 0.0116	0.9847 ± 0.0056	0.7584 ± 0.0368
Q49	0.7573 ± 0.0088	0.6128 ± 0.0226	0.6408 ± 0.0266	0.6161 ± 0.0119	0.9849 ± 0.0032	0.7595 ± 0.0365
UNIQUE	0.7584 ± 0.0090	0.6128 ± 0.0226	0.6408 ± 0.0266	0.6174 ± 0.0125	0.9857 ± 0.0046	0.7569 ± 0.0401

4.6.4. Full-Test Representativeness

To assess whether the primary 750-sample evaluation provided stable QAE performance estimates, the selected QAE was also evaluated on the full official UNSW-NB15 test partition after model selection. The resulting F1-score, AUPRC, and AUROC values were closely aligned with those obtained from the original 750-sample evaluation (Table 14). This agreement indicates that, for the selected QAE, the performance estimates obtained from the primary test subset remained largely consistent when evaluation was extended to the complete official test set.

Table 14. QAE performance on the primary 750-sample test subset and the full official UNSW-NB15 test partition.

Evaluation Set	F1	AUPRC	AUROC
Primary 750-sample subset	0.7576 ± 0.0088	0.6128 ± 0.0226	0.6408 ± 0.0266
Full official test partition	0.7569 ± 0.0066	0.6052 ± 0.0156	0.6359 ± 0.0180

4.6.5. Prior-Sensitivity Analysis

This analysis examined whether the results of the Bayesian comparisons depended on the choice of prior. The same QAE–classical model comparisons were recalculated using alternative priors (Table 15). None of the comparisons changed direction; in other words, a difference favoring the QAE did not switch to favoring the classical model, and vice versa.

Table 15. Prior-sensitivity summary for the paired Bayesian QAE–classical comparisons.

Prior	Specification	Sign Reversals vs. Reference Prior
Weakly informative NIG	$\kappa_0 = 0.10$, $\alpha_0 = 2$, $\beta_0 = 0.01$	0
Stronger centered NIG	$\kappa_0 = 1.00$, $\alpha_0 = 2$, $\beta_0 = 0.01$	0

4.7. Computational Cost

Computational cost differed substantially across the evaluated models. Under statevector simulation, the QAE required considerably longer training times than the classical methods, and its test-set inference time was also higher than that of all classical baselines (Table 16). Among the classical methods, OCSVM, LOF, and IF had the shortest training times, while CAE, ECOD, and COPOD required somewhat longer training but still remained far below the QAE.

Table 16. Computational cost of the QAE and classical anomaly detection models.

Model	Training Time (s)	Test Inference Time (s)
QAE	675.62 ± 127.75	8.938 ± 0.063
CAE	3.251 ± 0.013	0.00496 ± 0.00016
IF	0.156 ± 0.001	0.01101 ± 0.00016
LOF	0.0248 ± 0.0005	0.01089 ± 0.00031
OCSVM	0.0134 ± 0.0001	0.00511 ± 0.00009
HBOS	0.0443 ± 0.0004	0.00483 ± 0.00005
ECOD	3.410 ± 0.007	0.01329 ± 0.00006
COPOD	3.412 ± 0.020	0.01339 ± 0.00014

5. Discussion

The primary contribution of this study is the evaluation of QAE performance in network anomaly detection within a controlled, multidimensional, and standardized benchmark protocol, rather than relying on a single performance metric or an isolated classical

comparator. Within the primary matched benchmark, standardized experimental conditions were maintained through shared data partitions, uniform preprocessing pipelines (aside from QAE-specific angle encoding), predefined model selection procedures, and decision-threshold calibration rules. Conducting experiments across five independent random seeds, multiple training set sizes, and two distinct feature budgets reduced the dependence of the findings on a single data split or problem scale. Rather than indicating unconditional superiority or overall failure for the QAE, the results show that algorithmic performance is closely tied to the target metric and operational regime.

The performance profile of the QAE illustrates a critical distinction between threshold-independent score ranking quality and threshold-dependent operational decision performance. The QAE's competitive relative performance in AUPRC and AUROC on UNSW-NB15 supports the ability of the model to establish a continuous anomaly score ranking that separates nominal traffic from attack patterns. Conversely, empirical findings for the F1-score and low-FPR operating points show that global score discrimination does not automatically translate into effective operational classification. The marked reduction in QAE recall when tightening false-alarm constraints indicates that global ranking performance does not yield comparable attack-detection performance in the low-FPR operating region. The observation that LOF provides a more favorable recall–FPR balance under the evaluated low-FPR conditions underscores that anomaly detection model selection should not be based solely on global AUROC or AUPRC values. In domains such as network security where false alarms carry substantial operational costs, model performance at the intended operating points should be evaluated directly [61].

The threshold-sensitivity analysis showed that the QAE's threshold-dependent performance changed only slightly across the Q99, Q49, and UNIQUE strategies, suggesting that the main results were not strongly affected by the resolution of the quantile grid used for calibration. A broader tuning analysis for the CAE also produced only limited changes in performance: the small average improvement in AUPRC had a credible interval that included zero, while AUROC remained essentially unchanged. This suggests that the overall QAE–CAE comparison was not mainly driven by the narrower tuning space initially used for the CAE. The less-constrained classical reference analysis likewise showed that larger training sets and the full available feature space did not benefit all classical methods in the same way. Some methods improved, particularly on ranking-based metrics, whereas others showed only modest gains or even declines on certain measures. The constrained training sizes and 5- and 8-feature settings used in the matched benchmark therefore do not appear to have systematically suppressed the performance of all classical models. At the same time, because the reference analysis changed training size, feature space, and evaluation scope together, the observed differences should not be attributed to any single factor, and this analysis is best viewed as a complementary reference rather than a replacement for the matched benchmark. The PCA sensitivity analysis showed a different pattern: QAE performance was sensitive to feature representation, with both AUPRC and AUROC decreasing when the variance-based representation was replaced by PCA, particularly for AUROC. Finally, the paired Bayesian comparisons showed that relative model performance depended on the evaluation metric, while the direction of the comparisons remained unchanged under alternative prior specifications. Taken together, these analyses suggest that the main comparative conclusions were relatively stable with respect to threshold-grid resolution, CAE tuning breadth, and Bayesian prior choice, while feature representation remained an important factor shaping QAE performance.

QAE architecture-evaluation results showed that circuit-design choices can materially affect validation performance within the evaluated design space. The Angle-RY + Real-Amplitudes Ring configuration at depth $L = 2$ achieved the highest mean validation-

selection AUPRC and was therefore selected as the nominal top-ranked candidate. However, the architecture-stability analysis showed that its advantage over the two closest alternatives was small and was not consistently preserved in the leave-one-seed-out evaluations. The selected configuration should therefore be interpreted as the highest-ranked architecture within the predefined search space rather than as a clearly superior circuit design. At the same time, all $L = 2$ configurations outperformed their corresponding ($L = 1$) counterparts in validation performance, suggesting that the additional variational layer was beneficial within the shallow circuits examined here. Training dynamics also showed stable convergence in most conditions, with no clear evidence of near-zero gradients or optimization failure in the 5- and 8-qubit circuits with $L \leq 2$. However, these small-scale observations do not constitute a scaling-based test for barren plateaus, and the risk remains relevant as qubit count and circuit depth increase [69].

The independent cross-dataset evaluation on TON_IoT showed that model performance was sensitive to the dataset being used. The relative performance pattern observed on UNSW-NB15 was not preserved in the same way on TON_IoT, where LOF achieved the strongest overall results in terms of F1-score, AUPRC, and AUROC. The QAE continued to produce very high recall on TON_IoT, but this came with a substantially elevated false-positive rate. These findings show that the QAE architecture selected on UNSW-NB15 could be transferred to a second network-security dataset, but its relative performance and operational behavior were not dataset-independent. This reinforces the importance of evaluating QAE-based anomaly detection across different network environments and attack distributions rather than drawing conclusions from a single benchmark.

The reduced QAE recall observed at stringent low-FPR operating points should not be attributed to a single factor on the basis of the present experiments. The architecture evaluation showed that circuit design can affect QAE performance, but the low-FPR analysis was conducted only for the selected architecture and therefore does not isolate the contribution of architecture choice to low-FPR behavior. Similarly, the PCA sensitivity experiment demonstrated that changing the feature representation affected threshold-independent QAE discrimination, but it was not designed to quantify the specific effect of feature representation on recall at fixed low-FPR targets. The cross-dataset results further showed substantial changes in relative and threshold-dependent behavior between UNSW-NB15 and TON_IoT, indicating that dataset characteristics and anomaly distributions also influence operational performance. Consequently, the observed low-FPR recall degradation is best interpreted as a property of the evaluated QAE pipeline under the present architecture, feature representation, and dataset conditions, rather than as evidence of a single dominant cause. Isolating the individual contributions of these factors would require a dedicated factorial experiment in which architecture, feature representation, and dataset conditions are varied independently.

Finite-shot and synthetic-noise experiments showed that the QAE's anomaly-score ranking remained relatively stable, whereas its threshold-based operational decisions were more sensitive to noise. When the decision threshold was recalibrated for each noisy condition, F1-score and FPR remained broadly similar to the ideal setting. However, when the threshold calibrated under ideal execution was applied unchanged to noisy conditions, the false-positive rate increased substantially. This indicates that noise does not necessarily destroy the model's ability to rank anomalous samples, but it can shift score distributions enough to make a previously calibrated decision threshold unreliable.

The additional evaluation on the full test set showed that the QAE's F1-score, AUPRC, and AUROC obtained from the primary 750-sample test subset remained closely aligned with the results on the complete official UNSW-NB15 test partition of 82,332 samples. This agreement suggests that using the primary test subset did not materially alter the estimated

QAE performance. From a practical deployment perspective, the computational-cost results do not indicate an efficiency advantage for the QAE under the evaluated conditions. Under CPU-based statevector simulation, the QAE required substantially longer training and inference times than all classical baselines. This overhead is particularly relevant because the QAE did not consistently outperform the best-performing classical methods across the evaluated metrics, and its attack recall decreased markedly under stringent low-FPR constraints. However, the reported runtimes reflect classical simulation of the quantum circuit rather than execution on physical quantum hardware and therefore should not be interpreted as evidence either for or against a future hardware-level computational advantage. Within the scope of the present benchmark, the additional computational cost is therefore not offset by a demonstrated practical performance or efficiency benefit. This interpretation is consistent with recent guidance emphasizing that claims of practical QML advantage should be evaluated against well-tuned classical baselines with explicit consideration of computational resources and experimental conditions [9].

The controlled multi-experiment design, the use of two independent network-security datasets, and the evaluation across different training sizes, feature budgets, and sensitivity analyses reduce the likelihood that the findings are driven by a single experimental setting. Even so, the results should be interpreted within the scope of the present study. Although UNSW-NB15 and TON_IoT represent different traffic characteristics and attack distributions, evaluation on two datasets is not sufficient to establish broad dataset-independent generalization. The TON_IoT experiment should therefore be interpreted as an independent validation of whether the QAE architecture selected on UNSW-NB15 can be transferred to a second network-security dataset without additional architecture selection. The change in the relative performance of the evaluated methods across the two datasets further indicates that performance is influenced by dataset-specific characteristics. Validation on additional network-security datasets will therefore be important for assessing broader generalizability. Likewise, the five independent random seeds offer evidence of stability across multiple data partitions, while a larger number of independent repetitions could further refine the uncertainty estimates. Finally, the computational cost of quantum simulation limited the experiments to a maximum of 8 qubits and shallow circuit depths. Future work could therefore examine additional network-security datasets, larger circuit configurations, alternative latent-to-trash qubit ratios, training strategies designed specifically for the low-FPR region, and implementation on physical quantum processors.

6. Conclusions

This study compared a trash-qubit-based QAE with classical anomaly-detection methods representing different modeling approaches within a controlled benchmark framework. The results showed that the QAE did not consistently outperform the classical baselines; instead, its relative performance depended on the evaluation metric, dataset, and operating point. On UNSW-NB15, the QAE remained competitive in continuous-score discrimination, while its attack-detection performance became clearly limited under low-FPR conditions. The evaluation on TON_IoT further showed that relative model performance was dataset-dependent. Because the cross-dataset assessment is limited to UNSW-NB15 and TON_IoT, these results should not be interpreted as establishing broad dataset-independent generalization; validation on additional network-security datasets remains necessary.

The architecture-selection analysis identified the Angle-RY + Real-Amplitudes Ring configuration at depth $L = 2$ as the highest-ranked design by mean validation-selection AUPRC, although its advantage over the closest alternatives was limited. Finite-shot and synthetic-noise experiments also showed that anomaly-score rankings were largely

preserved, whereas decision thresholds were more sensitive to changes in the execution environment.

Overall, the findings suggest that the evaluated shallow QAE framework can produce informative continuous anomaly scores under benign-only one-class training, but turning these scores into reliable operational decisions at low false-alarm rates remains a key challenge. The controlled and reproducible evaluation framework presented here provides a useful basis for future studies of QAE-based network anomaly detection.

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Data Availability Statement: The UNSW-NB15 dataset used in this study is publicly available from the University of New South Wales (UNSW) at the official dataset repository. The experiments used the predefined UNSW_NB15_training-set.csv and UNSW_NB15_testing-set.csv partitions described in the dataset documentation [30]. The TON_IoT network dataset used for the cross-dataset evaluation is also publicly available from UNSW Canberra Cyber; this study used the train_test_network.csv file from the network dataset collection [39]. Upon acceptance, we will release the code under an open-source license on GitHub Desktop version 3.5.8 (x64) and archive a snapshot on Zenodo with a DOI.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AUPRC	Area Under the Precision–Recall Curve
AUROC	Area Under the Receiver Operating Characteristic Curve
CAE	Classical Autoencoder
CDF	Cumulative Distribution Function
CNOT	Controlled-NOT
COPOD	Copula-Based Outlier Detection
CPU	Central Processing Unit
CrI	Credible Interval
DeepSVDD	Deep Support Vector Data Description
ECOD	Empirical Cumulative Distribution-Based Outlier Detection
EE	Elliptic Envelope
FPR	False-Positive Rate
HBOS	Histogram-Based Outlier Score
IF	Isolation Forest
LOF	Local Outlier Factor
LR	Learning Rate
MCC	Matthews Correlation Coefficient
MSE	Mean Squared Error
OCSVM	One-Class Support Vector Machine
QAE	Quantum Autoencoder
QML	Quantum Machine Learning
RBF	Radial Basis Function
ReLU	Rectified Linear Unit

ROPE	Region of Practical Equivalence
SD	Standard Deviation
TON_IoT	Telemetry, Operating Systems, and Network Internet of Things Dataset
UNSW-NB15	University of New South Wales Network-Based 2015 Dataset
VQC	Variational Quantum Circuit
VQOCC	Variational Quantum One-Class Classifier

Appendix A

Table A1. Performance of the Additional Classical Baselines on UNSW-NB15.

Model	F1	AUPRC	AUROC	Precision	Recall	FPR
QAE	0.7576 ± 0.0088	0.6128 ± 0.0226	0.6408 ± 0.0266	0.6167 ± 0.0116	0.9847 ± 0.0056	0.7584 ± 0.0368
DeepSVDD	0.7164 ± 0.0068	0.5499 ± 0.0612	0.4943 ± 0.0606	0.5659 ± 0.0186	0.9810 ± 0.0228	0.9312 ± 0.0735
EE	0.8089 ± 0.0106	0.5980 ± 0.0193	0.6417 ± 0.0222	0.6861 ± 0.0124	0.9856 ± 0.0073	0.5534 ± 0.0290

Table A2. Performance of the Additional Classical Baselines on TON_IoT.

Model	F1	AUPRC	AUROC	Precision	Recall	FPR
QAE	0.7296 ± 0.0076	0.6396 ± 0.0245	0.5603 ± 0.0185	0.5774 ± 0.0106	0.9960 ± 0.0065	0.9085 ± 0.0319
DeepSVDD	0.7219 ± 0.0169	0.5867 ± 0.0421	0.4756 ± 0.0826	0.5736 ± 0.0254	0.9846 ± 0.0121	0.9223 ± 0.0705
EE	0.7535 ± 0.0198	0.5977 ± 0.0270	0.5375 ± 0.0526	0.6258 ± 0.0406	0.9680 ± 0.0228	0.7480 ± 0.1187

Table A3. Paired QAE Comparisons with the Additional Baselines on UNSW-NB15.

Metric	Comparator	Mean Δ	Median Δ	95% CrI	P(μ > 0|Data)	Seed Direction (QAE/Classical)
F1	DeepSVDD	+0.0412	+0.0409	[0.0337, 0.0487]	0.9999	5/0
	EE	−0.0513	−0.0498	[−0.0649, −0.0378]	0.0002	0/5
AUPRC	DeepSVDD	+0.0630	+0.0744	[−0.0024, 0.1283]	0.9723	4/1
	EE	+0.0148	+0.0131	[−0.0144, 0.0440]	0.8842	3/2
AUROC	DeepSVDD	+0.1465	+0.1357	[0.0728, 0.2202]	0.9974	5/0
	EE	−0.0009	+0.0033	[−0.0288, 0.0269]	0.4658	3/2

Table A4. Low-FPR performance under large-sample benign-only calibration on the full UNSW-NB15 test set.

Model	Recall @ 1%	Realized FPR @ 1%	Recall @ 5%	Realized FPR @ 5%	Recall @ 10%	Realized FPR @ 10%
QAE	0.0016 ± 0.0022	0.0093 ± 0.0005	0.0122 ± 0.0081	0.0537 ± 0.0063	0.0497 ± 0.0173	0.1193 ± 0.0171
CAE	0.0171 ± 0.0058	0.0202 ± 0.0074	0.0919 ± 0.0180	0.0926 ± 0.0256	0.2226 ± 0.0464	0.1951 ± 0.0344
IF	0.0038 ± 0.0016	0.0215 ± 0.0026	0.0495 ± 0.0064	0.0858 ± 0.0089	0.1035 ± 0.0174	0.1536 ± 0.0130
LOF	0.0150 ± 0.0009	0.0075 ± 0.0025	0.0412 ± 0.0059	0.0437 ± 0.0047	0.0961 ± 0.0241	0.0904 ± 0.0063
OCSVM	0.0094 ± 0.0005	0.0113 ± 0.0017	0.0290 ± 0.0024	0.0626 ± 0.0057	0.0291 ± 0.0024	0.0923 ± 0.0055
HBOS	0.0041 ± 0.0022	0.0145 ± 0.0033	0.0198 ± 0.0107	0.0385 ± 0.0115	0.1090 ± 0.0466	0.1573 ± 0.0193
ECOD	0.0050 ± 0.0007	0.0230 ± 0.0013	0.0307 ± 0.0048	0.0870 ± 0.0038	0.0573 ± 0.0077	0.1435 ± 0.0046
COPOD	0.0042 ± 0.0009	0.0223 ± 0.0012	0.0316 ± 0.0048	0.0848 ± 0.0054	0.0671 ± 0.0092	0.1667 ± 0.0107
DeepSVDD	0.0066 ± 0.0011	0.0107 ± 0.0028	0.1255 ± 0.1406	0.0577 ± 0.0172	0.1728 ± 0.1547	0.1066 ± 0.0212
EE	0.0013 ± 0.0004	0.0063 ± 0.0008	0.0018 ± 0.0006	0.0282 ± 0.0023	0.0821 ± 0.0395	0.1214 ± 0.0095

Table A5. Additional operational metrics at the 1% nominal FPR target under large-sample benign-only calibration on the full UNSW-NB15 test partition.

Model	F1	Precision	Specificity	Balanced Accuracy	MCC
QAE	0.0032 ± 0.0043	0.0670 ± 0.0833	0.9907 ± 0.0005	0.4962 ± 0.0009	−0.0601 ± 0.0131
CAE	0.0328 ± 0.0107	0.4422 ± 0.0402	0.9798 ± 0.0074	0.4985 ± 0.0010	−0.0191 ± 0.0082
IF	0.0074 ± 0.0032	0.1632 ± 0.0470	0.9785 ± 0.0026	0.4911 ± 0.0005	−0.0826 ± 0.0028
LOF	0.0294 ± 0.0017	0.7187 ± 0.0522	0.9925 ± 0.0025	0.5038 ± 0.0008	0.0358 ± 0.0095
OCSVM	0.0185 ± 0.0009	0.5078 ± 0.0235	0.9887 ± 0.0017	0.4991 ± 0.0006	−0.0089 ± 0.0053
HBOS	0.0081 ± 0.0043	0.2346 ± 0.0726	0.9855 ± 0.0033	0.4948 ± 0.0008	−0.0544 ± 0.0040
ECOD	0.0097 ± 0.0014	0.2005 ± 0.0320	0.9770 ± 0.0013	0.4910 ± 0.0006	−0.0790 ± 0.0048
COPOD	0.0083 ± 0.0017	0.1764 ± 0.0414	0.9777 ± 0.0012	0.4910 ± 0.0005	−0.0817 ± 0.0055
DeepSVDD	0.0129 ± 0.0021	0.4343 ± 0.0767	0.9893 ± 0.0028	0.4980 ± 0.0016	−0.0218 ± 0.0157
EE	0.0026 ± 0.0009	0.1793 ± 0.0424	0.9937 ± 0.0008	0.4975 ± 0.0003	−0.0425 ± 0.0038

Table A6. Additional operational metrics at the 5% nominal FPR target under large-sample benign-only calibration on the full UNSW-NB15 test partition.

Model	F1	Precision	Specificity	Balanced Accuracy	MCC
QAE	0.0226 ± 0.0148	0.1730 ± 0.1094	0.9463 ± 0.0063	0.4793 ± 0.0045	−0.1232 ± 0.0314
CAE	0.1499 ± 0.0264	0.4728 ± 0.0194	0.9074 ± 0.0256	0.4996 ± 0.0051	−0.0188 ± 0.0101
IF	0.0873 ± 0.0101	0.3977 ± 0.0207	0.9142 ± 0.0089	0.4819 ± 0.0026	−0.0758 ± 0.0094
LOF	0.0764 ± 0.0104	0.5326 ± 0.0212	0.9563 ± 0.0047	0.4988 ± 0.0016	−0.0067 ± 0.0083
OCSVM	0.0537 ± 0.0041	0.3625 ± 0.0048	0.9374 ± 0.0057	0.4832 ± 0.0017	−0.0812 ± 0.0051
HBOS	0.0346 ± 0.0174	0.2788 ± 0.0493	0.9615 ± 0.0115	0.4906 ± 0.0042	−0.0594 ± 0.0142
ECOD	0.0553 ± 0.0086	0.2951 ± 0.0343	0.9130 ± 0.0038	0.4718 ± 0.0018	−0.1225 ± 0.0108
COPOD	0.0570 ± 0.0086	0.3041 ± 0.0286	0.9152 ± 0.0054	0.4734 ± 0.0008	−0.1164 ± 0.0060
DeepSVDD	0.1690 ± 0.1760	0.4211 ± 0.2016	0.9423 ± 0.0172	0.5339 ± 0.0626	0.0313 ± 0.1563
EE	0.0034 ± 0.0011	0.0683 ± 0.0158	0.9718 ± 0.0023	0.4868 ± 0.0009	−0.1135 ± 0.0026

Table A7. Additional operational metrics at the 10% nominal FPR target under large-sample benign-only calibration on the full UNSW-NB15 test partition.

Model	F1	Precision	Specificity	Balanced Accuracy	MCC
QAE	0.0851 ± 0.0276	0.3134 ± 0.0622	0.8807 ± 0.0171	0.4652 ± 0.0046	−0.1316 ± 0.0221
CAE	0.3052 ± 0.0397	0.5418 ± 0.0243	0.8049 ± 0.0344	0.5138 ± 0.0156	0.0194 ± 0.0324
IF	0.1652 ± 0.0240	0.4347 ± 0.0208	0.8464 ± 0.0130	0.4749 ± 0.0027	−0.0804 ± 0.0112
LOF	0.1592 ± 0.0333	0.5236 ± 0.0374	0.9096 ± 0.0063	0.5028 ± 0.0096	−0.0027 ± 0.0266
OCSVM	0.0527 ± 0.0040	0.2788 ± 0.0123	0.9077 ± 0.0055	0.4684 ± 0.0021	−0.1349 ± 0.0067
HBOS	0.1703 ± 0.0671	0.4204 ± 0.1172	0.8427 ± 0.0193	0.4758 ± 0.0306	−0.0832 ± 0.0896
ECOD	0.0967 ± 0.0129	0.3211 ± 0.0322	0.8565 ± 0.0046	0.4569 ± 0.0027	−0.1467 ± 0.0134
COPOD	0.1106 ± 0.0142	0.3270 ± 0.0187	0.8333 ± 0.0107	0.4502 ± 0.0010	−0.1575 ± 0.0043
DeepSVDD	0.2203 ± 0.1770	0.4153 ± 0.2182	0.8934 ± 0.0212	0.5331 ± 0.0708	0.0189 ± 0.1732
EE	0.1344 ± 0.0553	0.4137 ± 0.0693	0.8786 ± 0.0095	0.4803 ± 0.0163	−0.0772 ± 0.0489

Table A8. Realized FPR with condition-level 95% Wilson confidence intervals.

Model	1% Target	5% Target	10% Target
QAE	0.0093 ± 0.0005 [0.0035–0.0183]	0.0537 ± 0.0063 [0.0186–0.0891]	0.1193 ± 0.0171 [0.0497–0.1695]
CAE	0.0202 ± 0.0074 [0.0043–0.0350]	0.0926 ± 0.0256 [0.0280–0.1387]	0.1951 ± 0.0344 [0.1306–0.2506]
IF	0.0215 ± 0.0026 [0.0115–0.0312]	0.0858 ± 0.0089 [0.0476–0.1315]	0.1536 ± 0.0130 [0.1151–0.1955]
LOF	0.0075 ± 0.0025 [0.0044–0.0159]	0.0437 ± 0.0047 [0.0301–0.0559]	0.0904 ± 0.0063 [0.0689–0.1200]
OCSVM	0.0113 ± 0.0017 [0.0082–0.0155]	0.0626 ± 0.0057 [0.0501–0.0769]	0.0923 ± 0.0055 [0.0757–0.1087]
HBOS	0.0145 ± 0.0033 [0.0026–0.0349]	0.0385 ± 0.0115 [0.0026–0.1500]	0.1573 ± 0.0193 [0.0997–0.2563]
ECOD	0.0230 ± 0.0013 [0.0133–0.0340]	0.0870 ± 0.0038 [0.0576–0.1145]	0.1435 ± 0.0046 [0.1037–0.1842]
COPOD	0.0223 ± 0.0012 [0.0130–0.0324]	0.0848 ± 0.0054 [0.0536–0.1151]	0.1667 ± 0.0107 [0.1122–0.2071]
DeepSVDD	0.0107 ± 0.0028 [0.0058–0.0186]	0.0577 ± 0.0172 [0.0267–0.1234]	0.1066 ± 0.0212 [0.0572–0.2346]
EE	0.0063 ± 0.0008 [0.0029–0.0143]	0.0282 ± 0.0023 [0.0225–0.0441]	0.1214 ± 0.0095 [0.0980–0.1636]

Table A9. Attack recall with condition-level 95% Wilson confidence intervals.

Model	1% Target	5% Target	10% Target
QAE	0.0016 ± 0.0022 [0.0000–0.0154]	0.0122 ± 0.0081 [0.0000–0.0483]	0.0497 ± 0.0173 [0.0000–0.1291]
CAE	0.0171 ± 0.0058 [0.0007–0.0488]	0.0919 ± 0.0180 [0.0044–0.3210]	0.2226 ± 0.0464 [0.0612–0.7946]
IF	0.0038 ± 0.0016 [0.0005–0.0091]	0.0495 ± 0.0064 [0.0123–0.0951]	0.1035 ± 0.0174 [0.0479–0.2020]
LOF	0.0150 ± 0.0009 [0.0126–0.0208]	0.0412 ± 0.0059 [0.0247–0.0562]	0.0961 ± 0.0241 [0.0331–0.3002]
OCSVM	0.0094 ± 0.0005 [0.0080–0.0111]	0.0290 ± 0.0024 [0.0235–0.0361]	0.0291 ± 0.0024 [0.0238–0.0361]
HBOS	0.0041 ± 0.0022 [0.0000–0.0125]	0.0198 ± 0.0107 [0.0000–0.1202]	0.1090 ± 0.0466 [0.0219–0.3323]
ECOD	0.0050 ± 0.0007 [0.0014–0.0088]	0.0307 ± 0.0048 [0.0137–0.0425]	0.0573 ± 0.0077 [0.0271–0.0791]
COPOD	0.0042 ± 0.0009 [0.0007–0.0081]	0.0316 ± 0.0048 [0.0128–0.0472]	0.0671 ± 0.0092 [0.0378–0.0833]
DeepSVDD	0.0066 ± 0.0011 [0.0033–0.0109]	0.1255 ± 0.1406 [0.0039–0.5545]	0.1728 ± 0.1547 [0.0039–0.6609]
EE	0.0013 ± 0.0004 [0.0000–0.0051]	0.0018 ± 0.0006 [0.0000–0.0054]	0.0821 ± 0.0395 [0.0415–0.3543]

Table A10. Score-rank stability and decision disagreement under finite-shot and synthetic-noise conditions.

Execution Condition	Spearman ρ vs. Ideal	Decision Disagreement Rate
Analytic simulation	1.0000 ± 0.0000	0.0000 ± 0.0000
256 shots	0.8977 ± 0.0162	0.0357 ± 0.0104
1024 shots	0.9161 ± 0.0074	0.0251 ± 0.0076
Mild noise, analytic	0.9671 ± 0.0528	0.0077 ± 0.0066
Moderate noise, analytic	0.9411 ± 0.0773	0.0124 ± 0.0056
Mild noise + 1024 shots	0.8891 ± 0.0457	0.0311 ± 0.0082

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